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Yoshida et al.

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(54) **IMAGE SELECTION APPARATUS, IMAGE SELECTION METHOD, AND NON-TRANSITORY COMPUTER-READABLE MEDIUM**

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G06V 20/40 (2022.01)

(Continued)

(52) **U.S. Cl.**

CPC **G06F 16/7328** (2019.01); **G06V 20/41** (2022.01); **G06V 20/48** (2022.01); **G06V 20/52** (2022.01); **G06V 10/945** (2022.01)

(58) **Field of Classification Search**

CPC **G06F 16/7328**; **G06V 20/52**; **G06V 20/41**; **G06V 20/48**; **G06V 10/945**

See application file for complete search history.

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Assistant Examiner — Dustin Bilodeau

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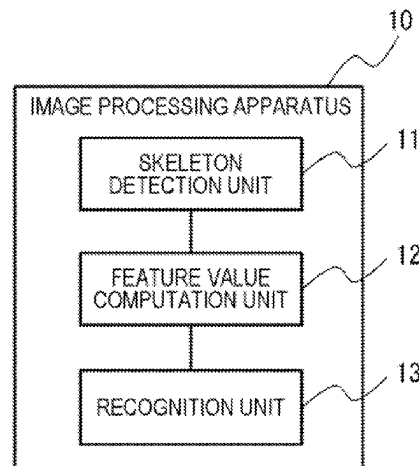
ABSTRACT

A query acquisition unit (610) acquires a query video. A query frame selection unit (620) selects a plurality of query frames from among a plurality of first frame images included in the query video. A video selection unit (630) selects a video by using the plurality of query frames. At this time, the video selection unit (630) uses references (1) and (2) below as a condition for selecting the video.

Reference (1) A condition that a “similar frame image whose degree of similarity to a query frame satisfies a first reference is present” is satisfied for at least two query frames.

Reference (2) When query frames associated with each of at least two similar frame images are arranged in the same order as that of the at least two similar frame images, the arrangement order coincides with an arrangement order of at least two query frames in a query video.

11 Claims, 47 Drawing Sheets



- (51) **Int. Cl.**
G06V 20/52 (2022.01)
G06V 10/94 (2022.01)

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FIG. 1

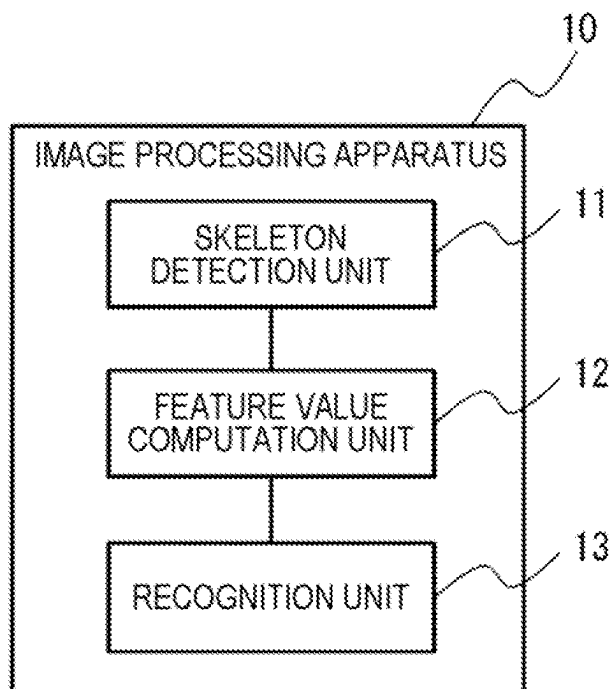


FIG. 2

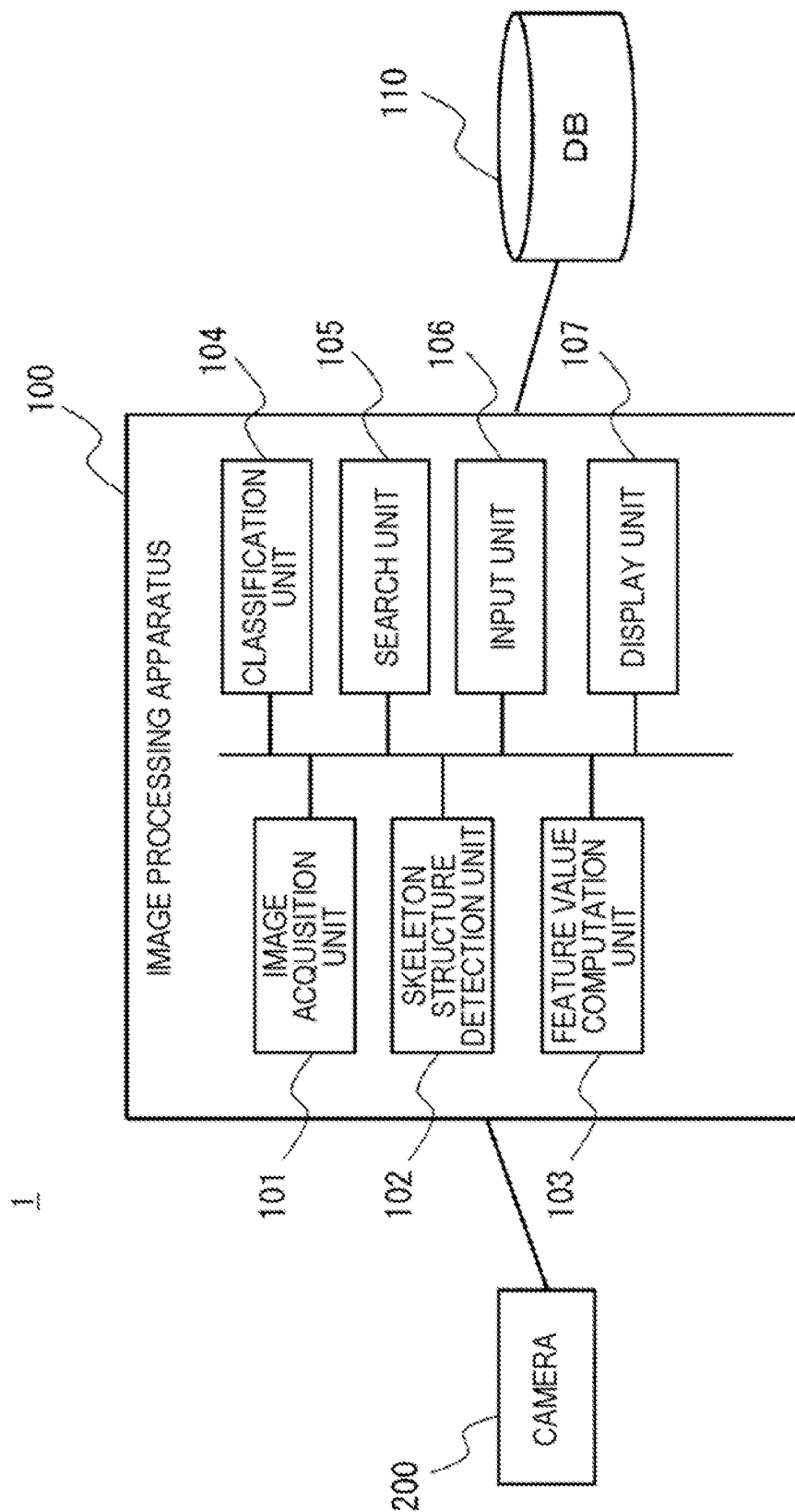


FIG. 3

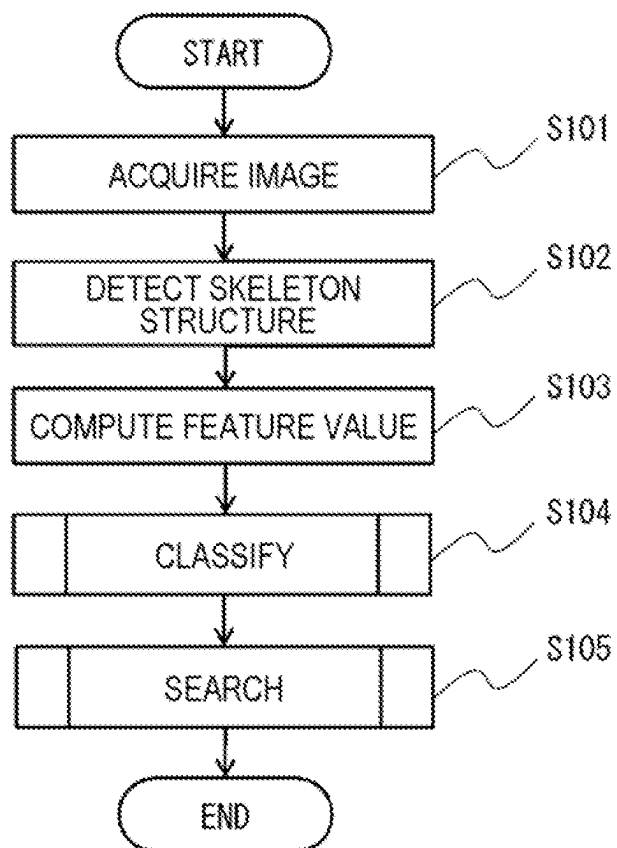


FIG. 4

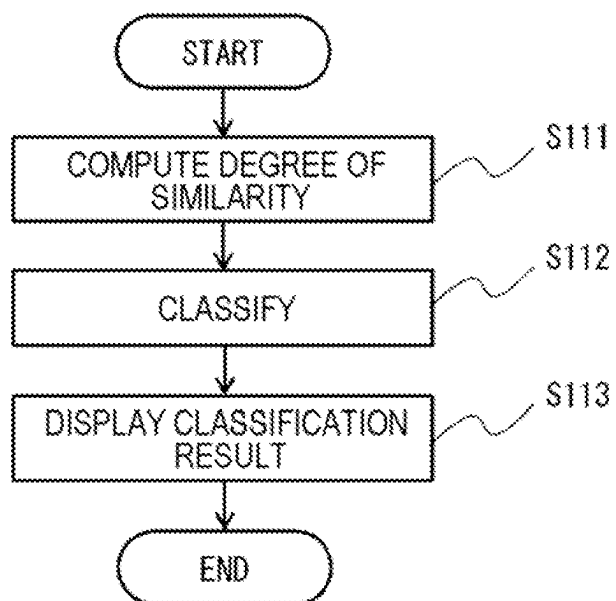
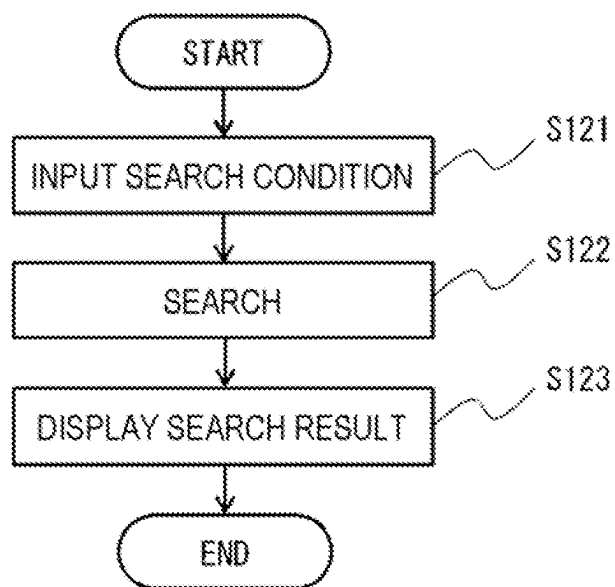


FIG. 5



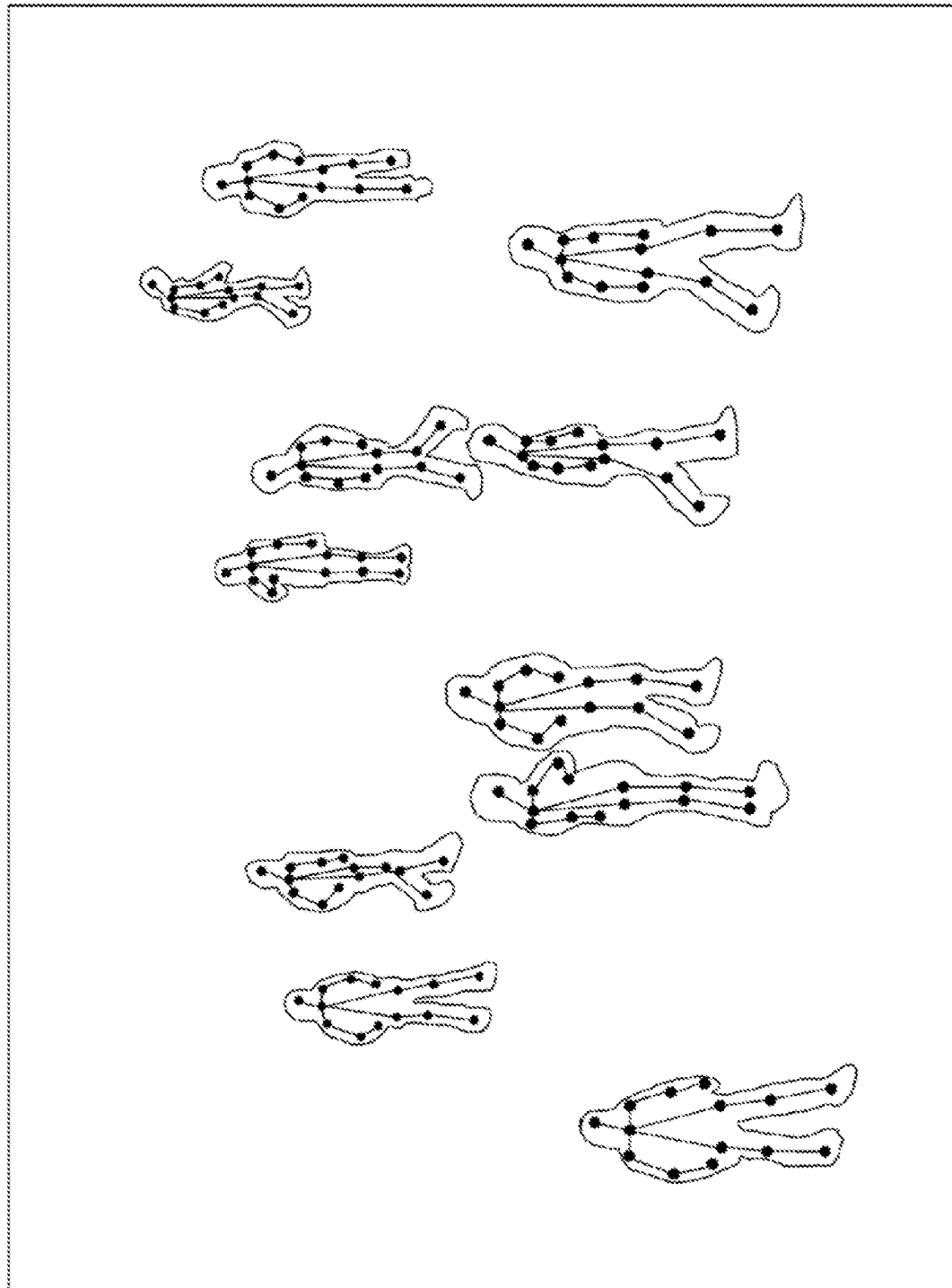


FIG. 6

FIG. 7

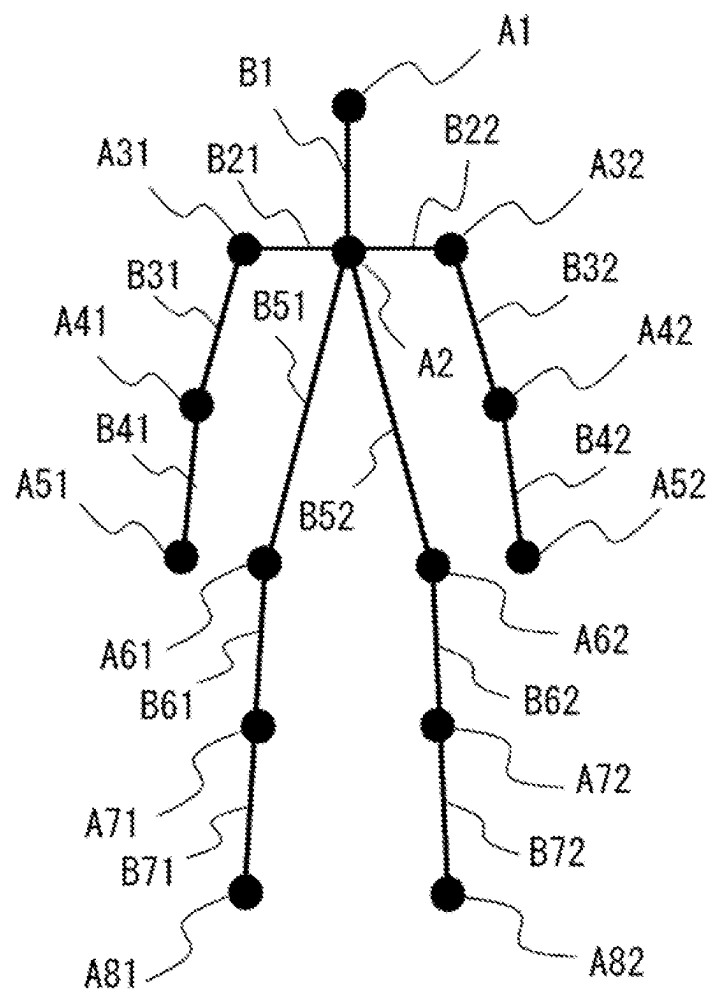
300

FIG. 8

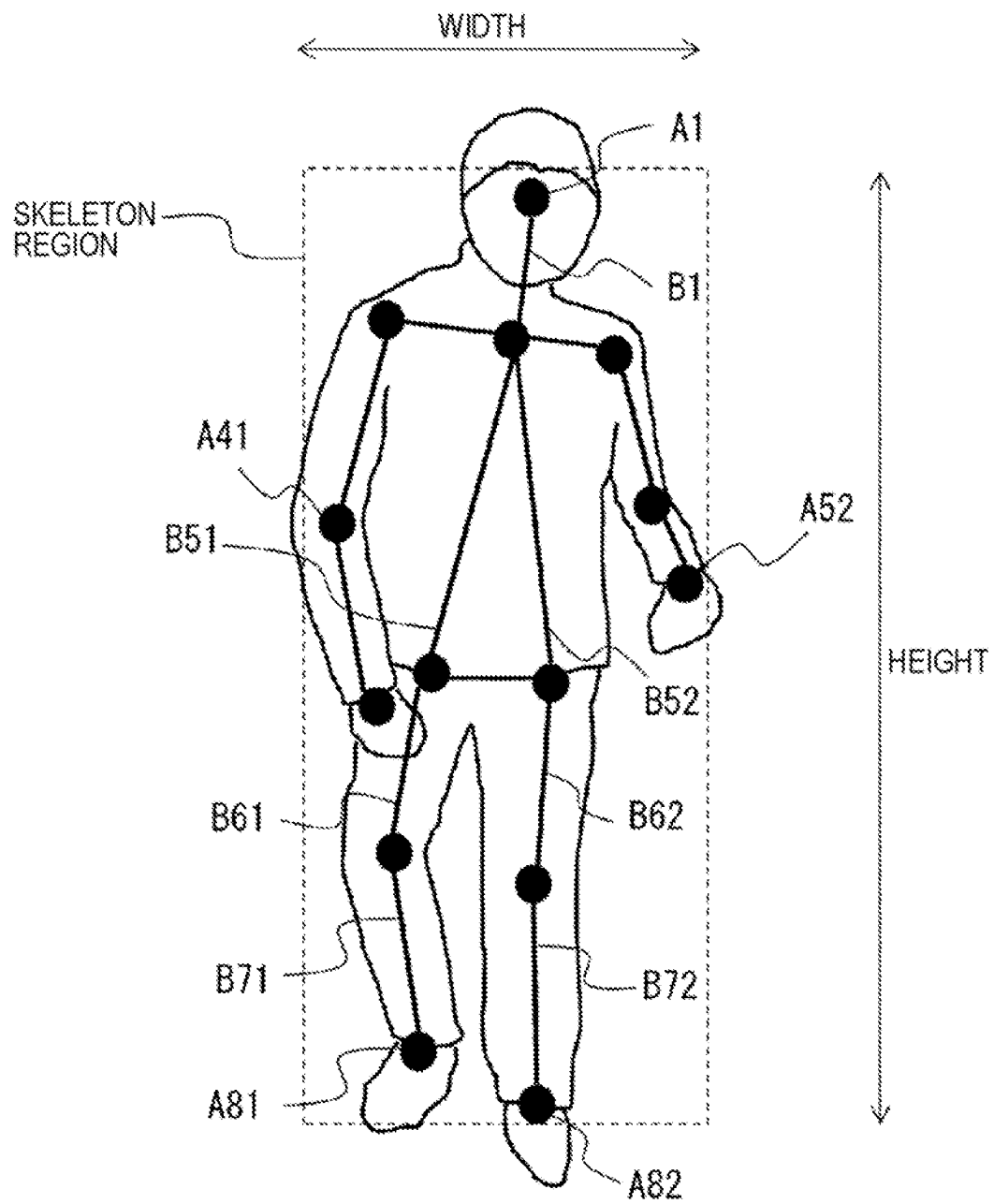


FIG. 9

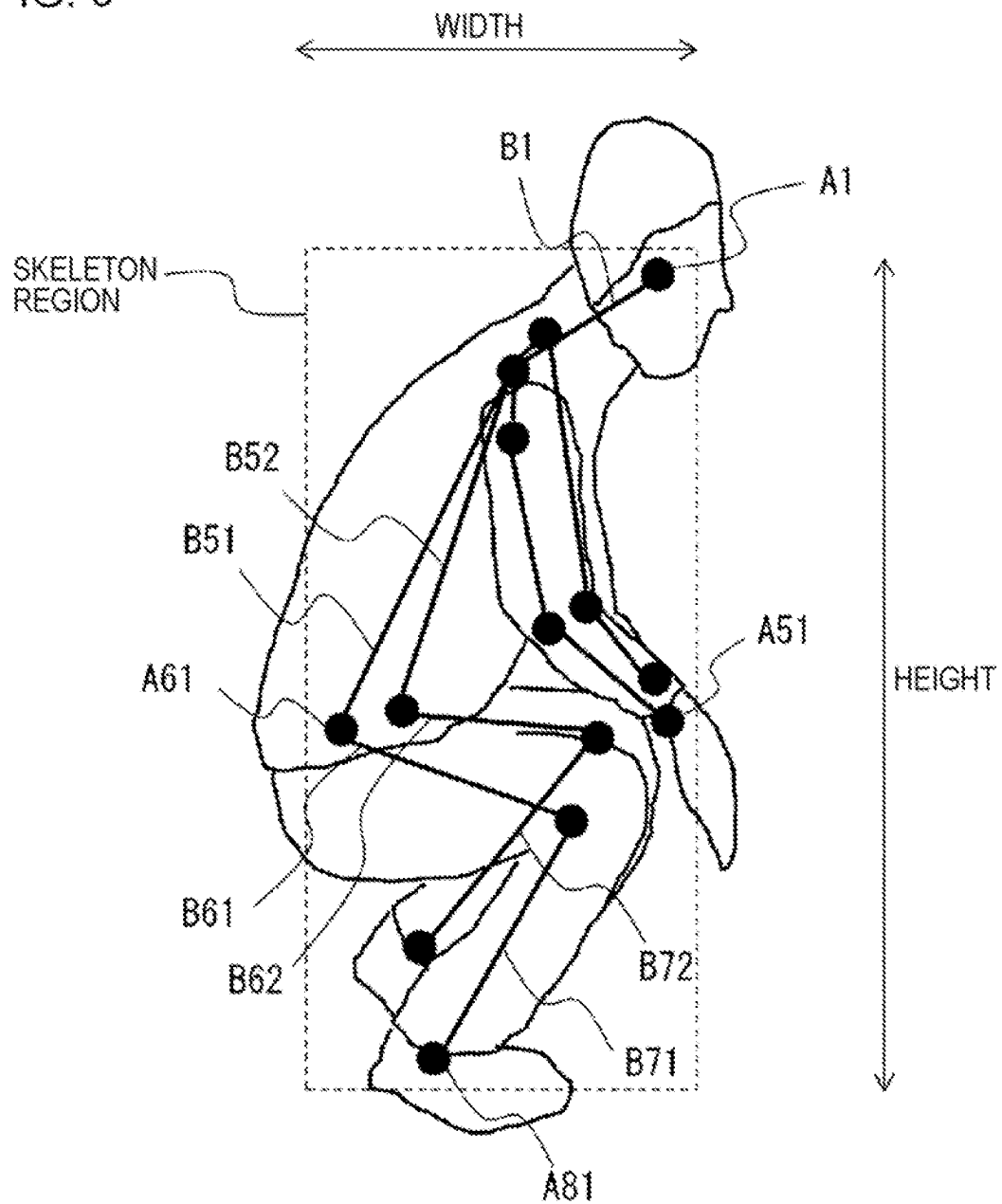
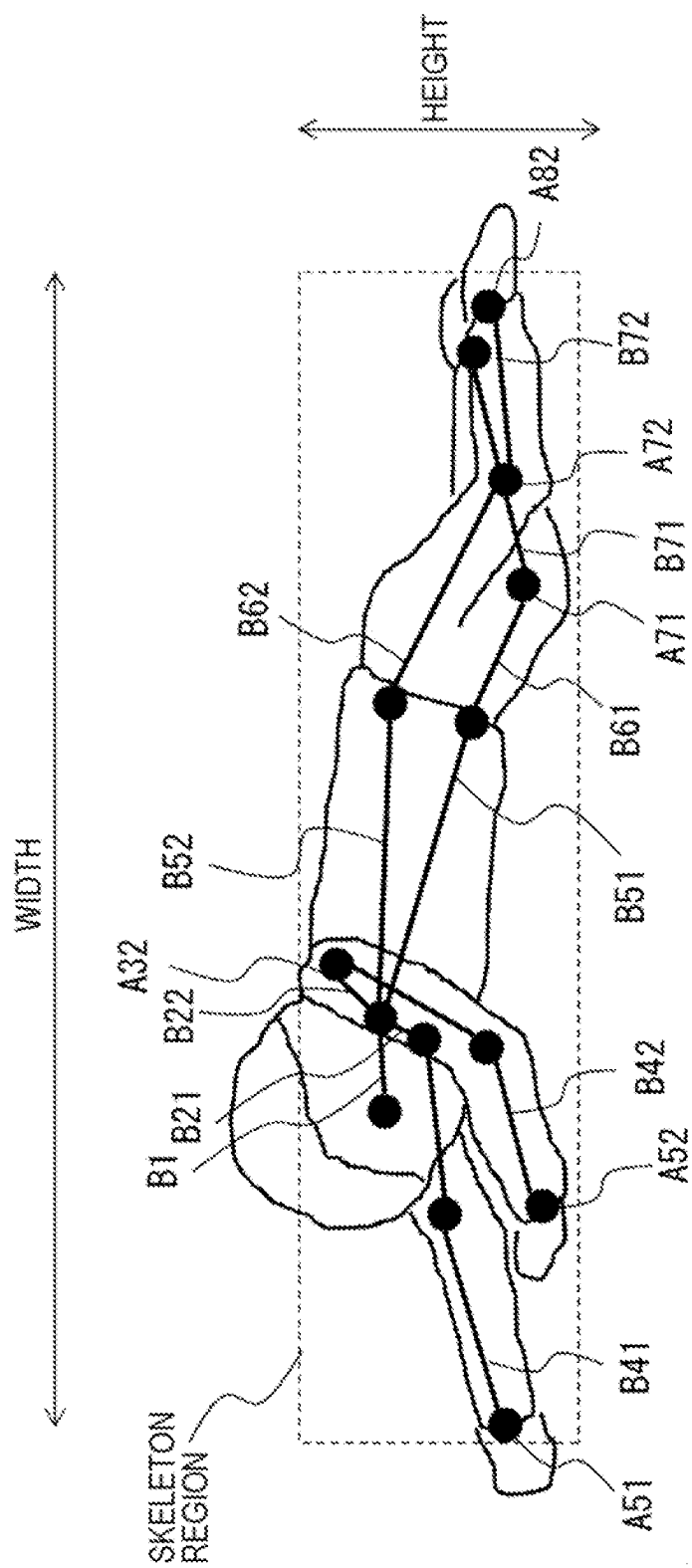


FIG. 10



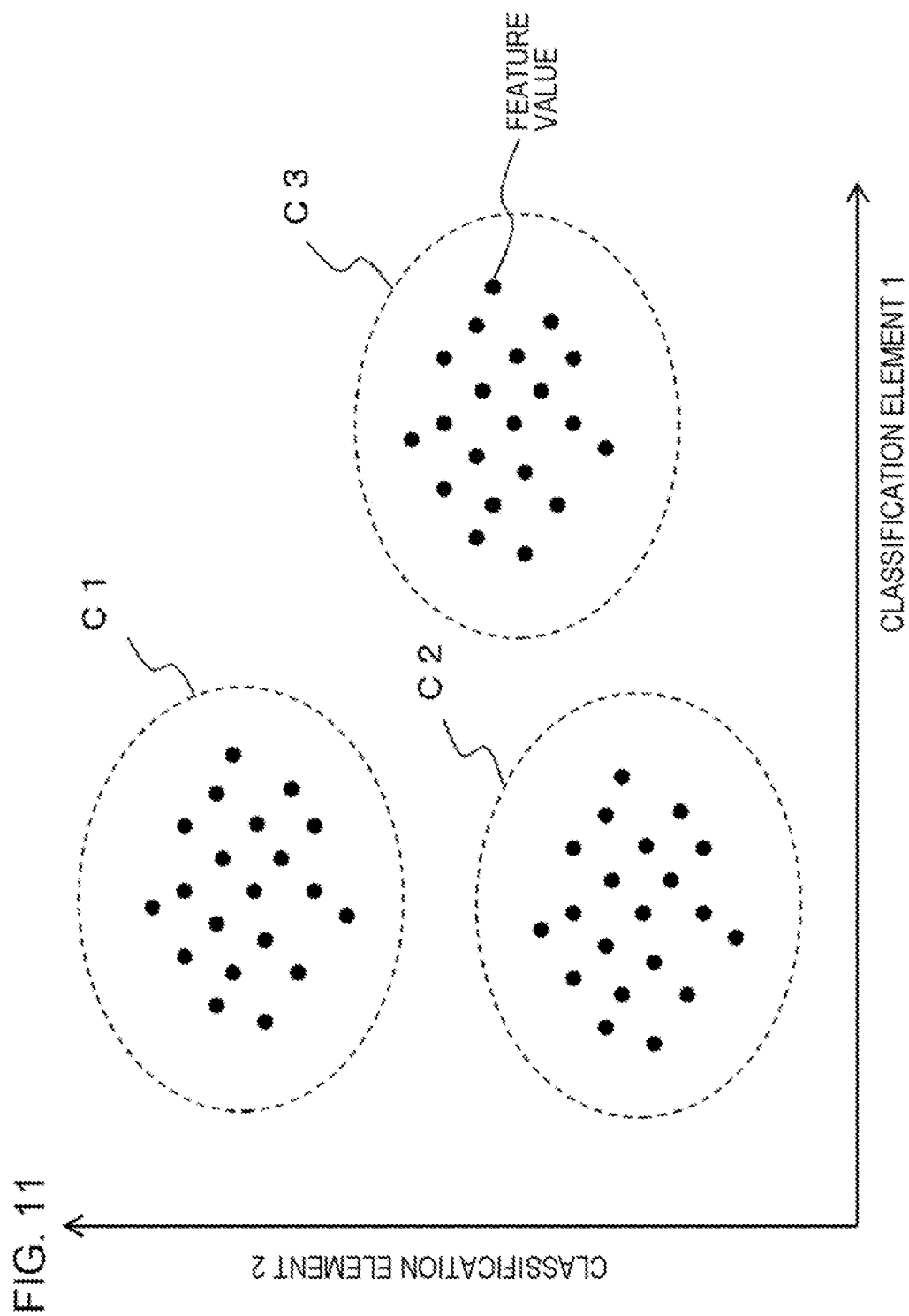
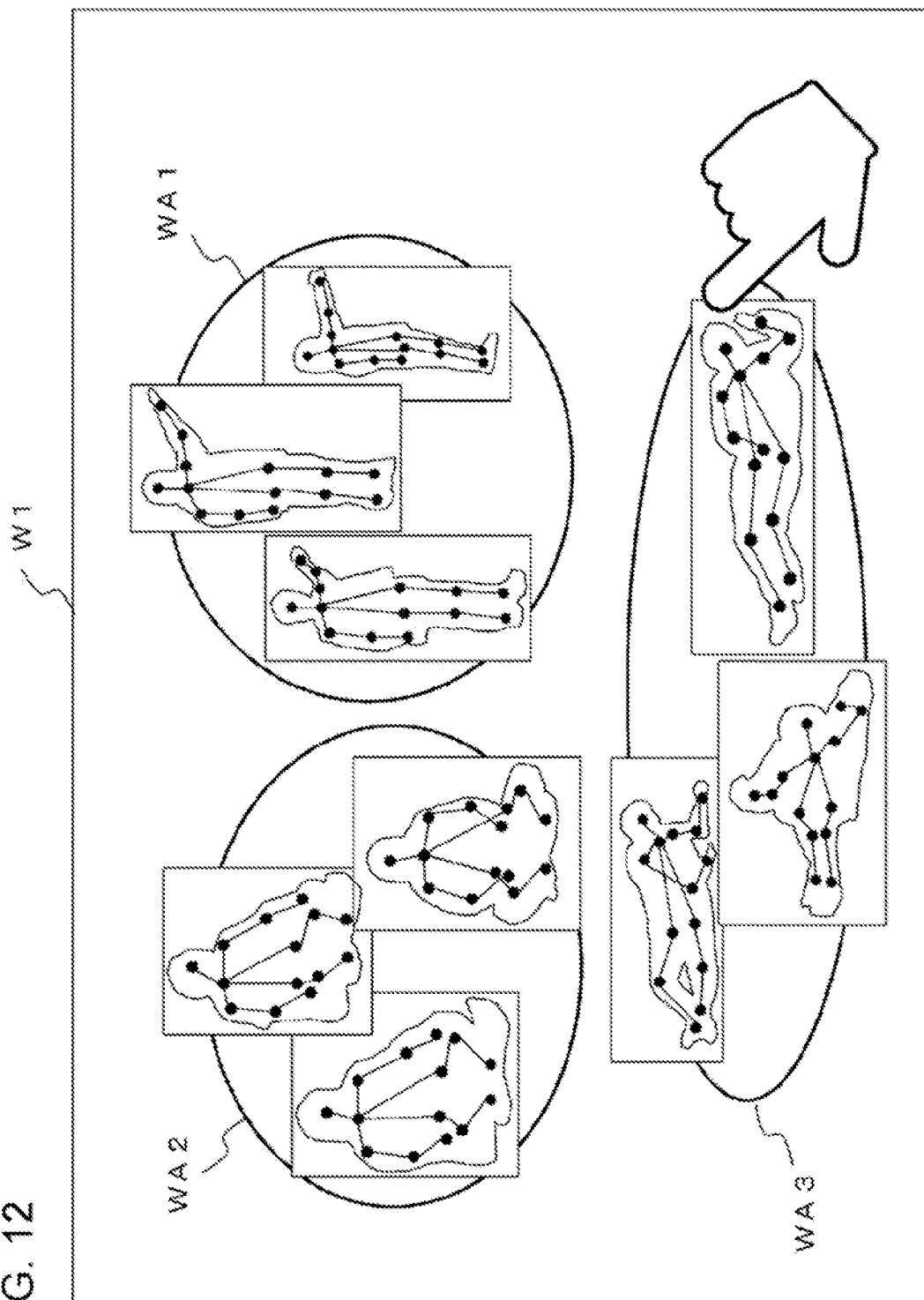


FIG. 12



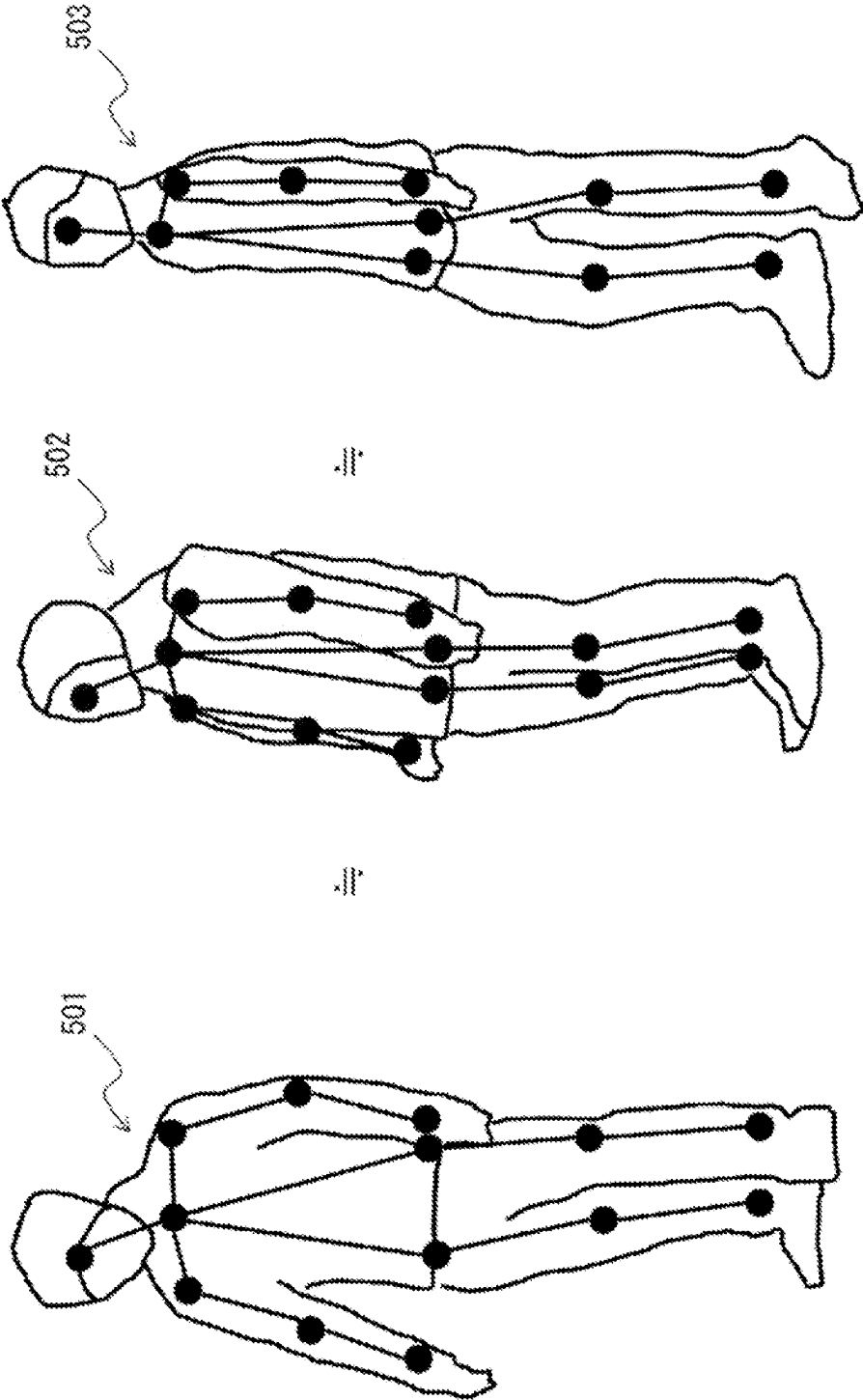


FIG. 13

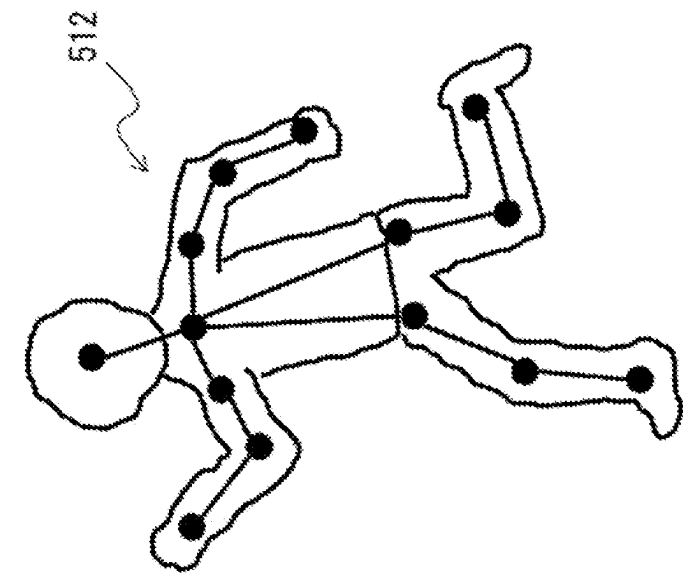


FIG. 14

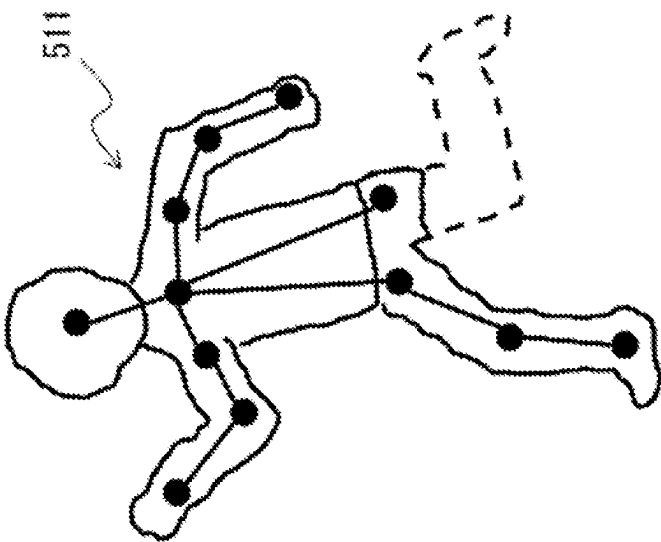
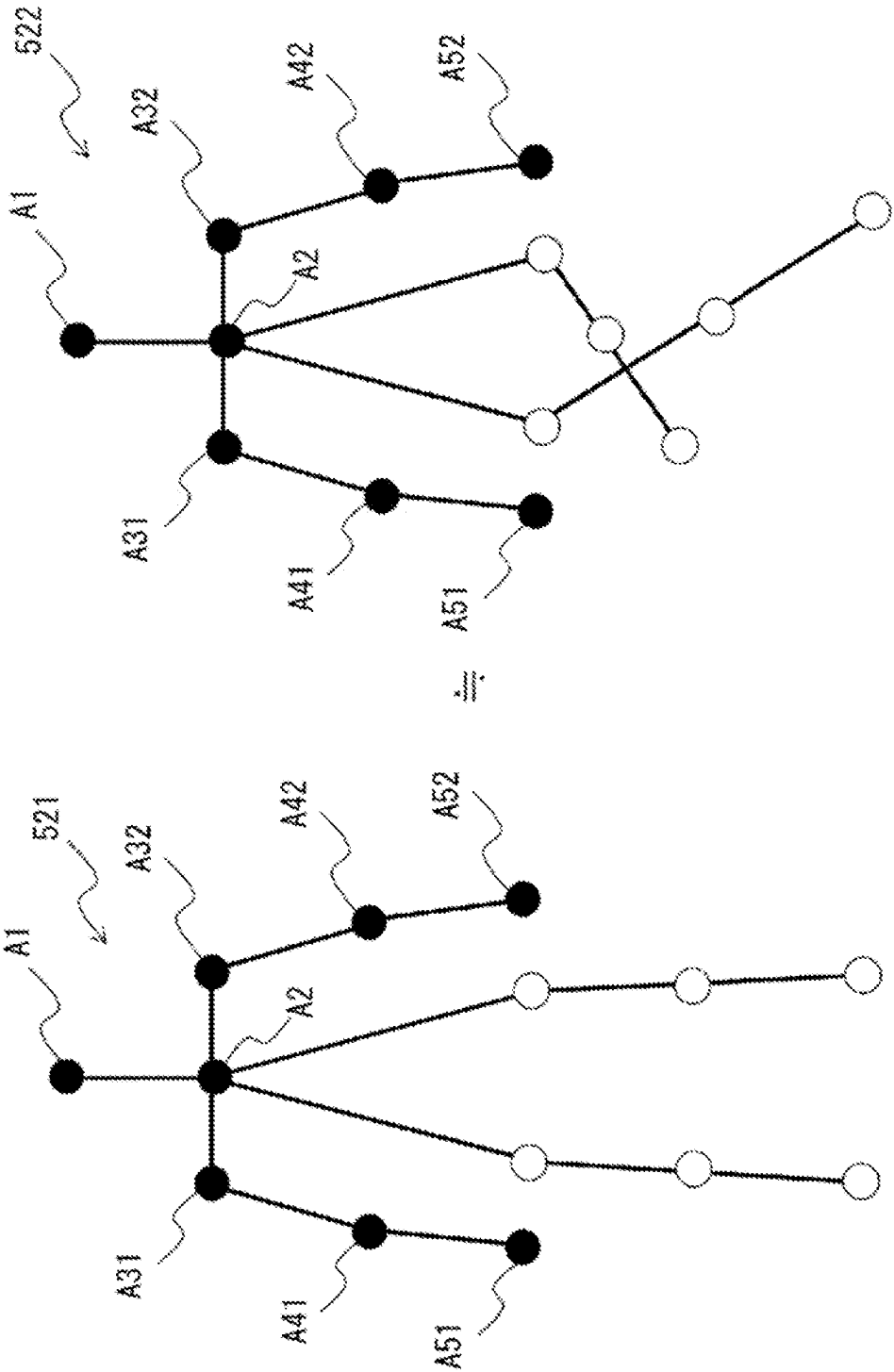
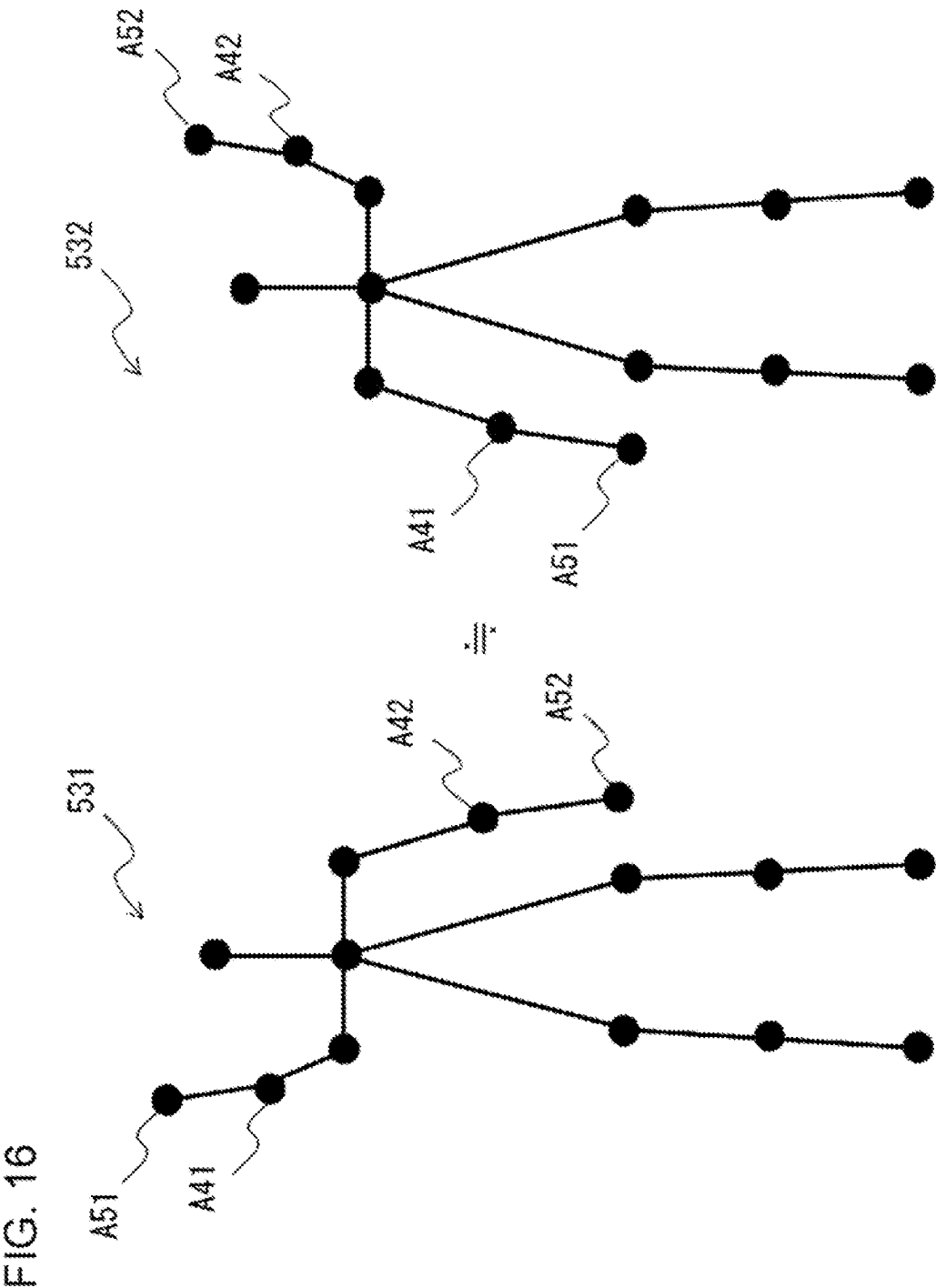


FIG. 15





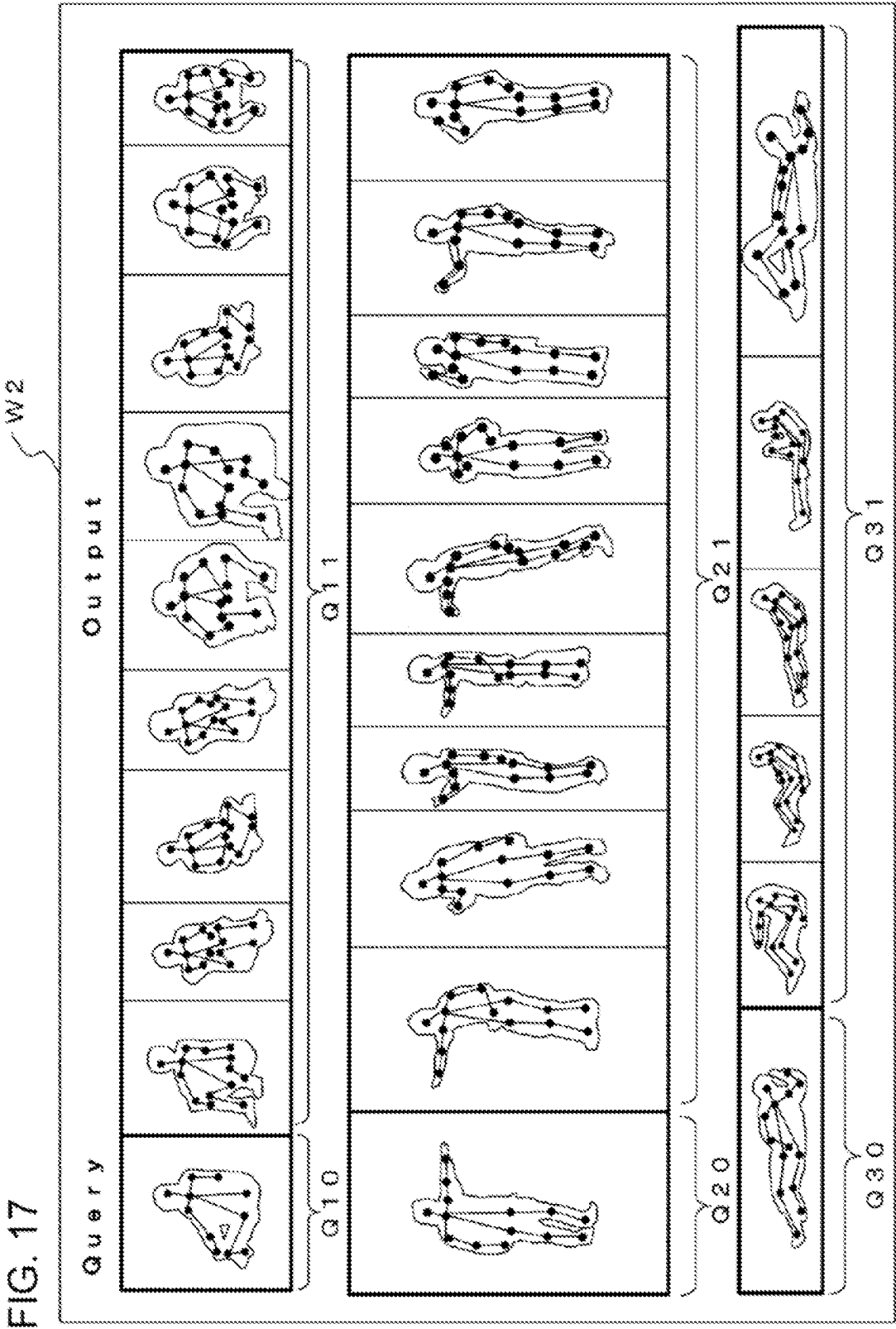


FIG. 18

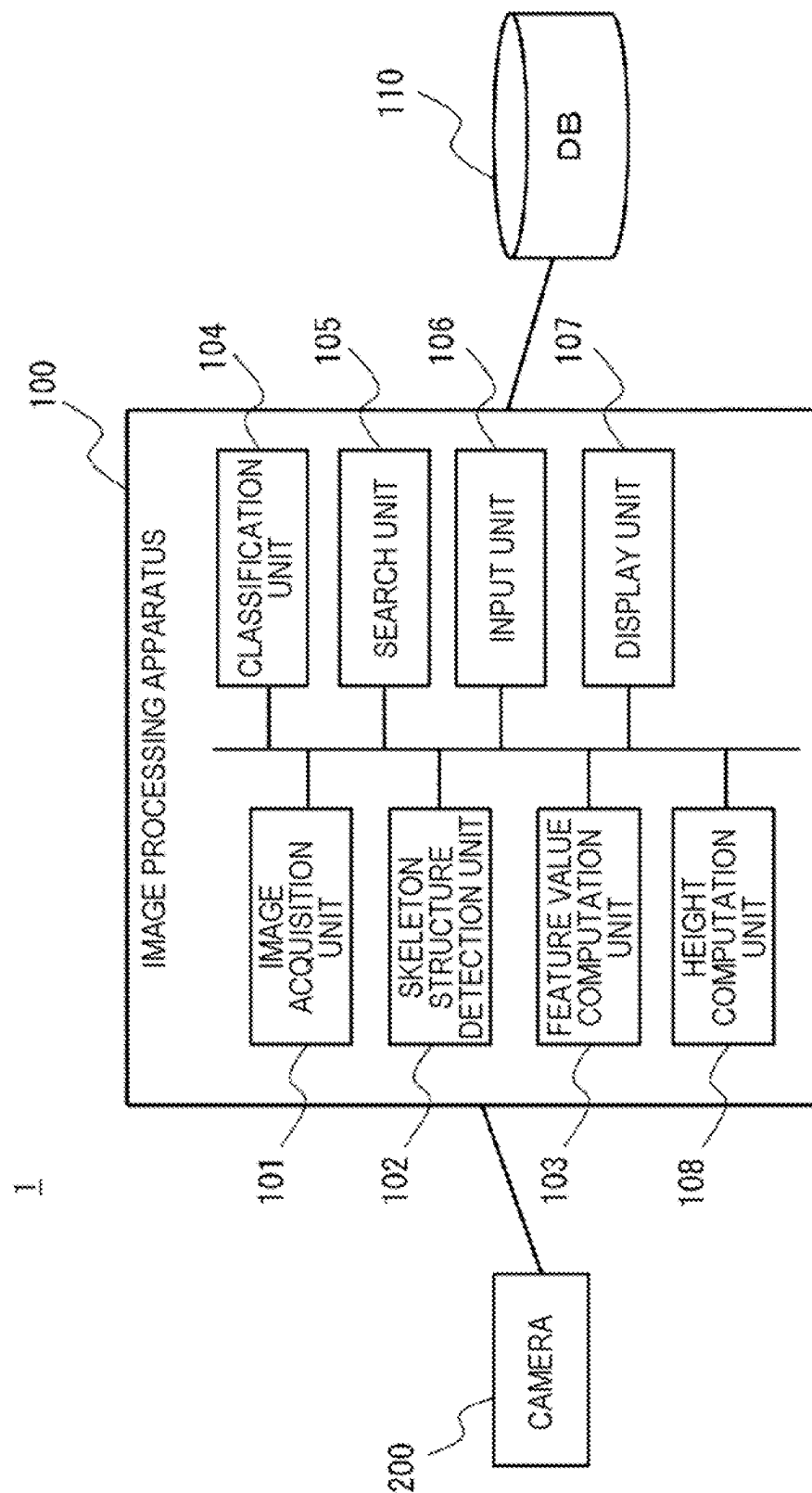


FIG. 19

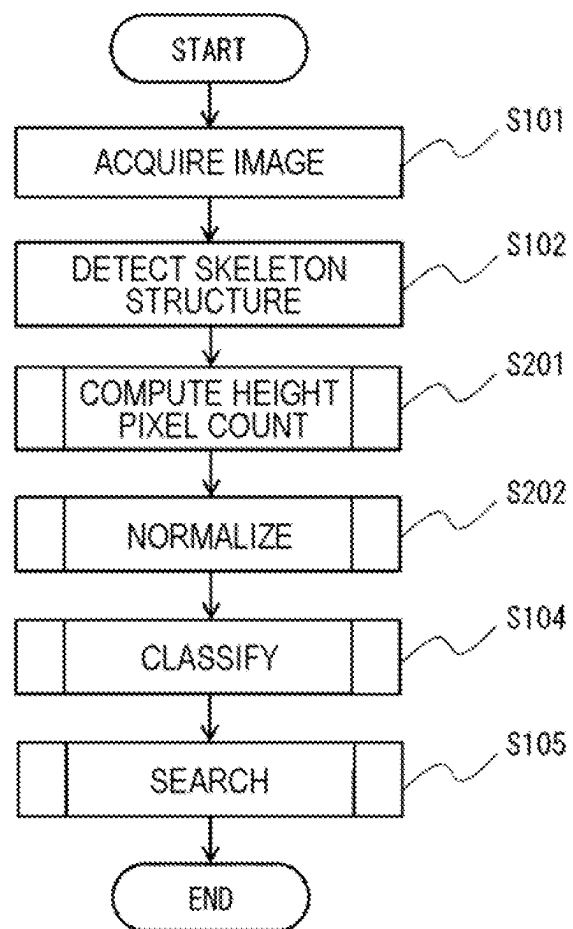


FIG. 20

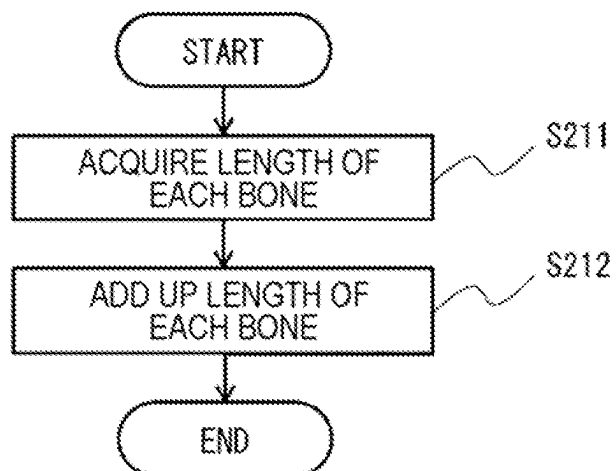


FIG. 21

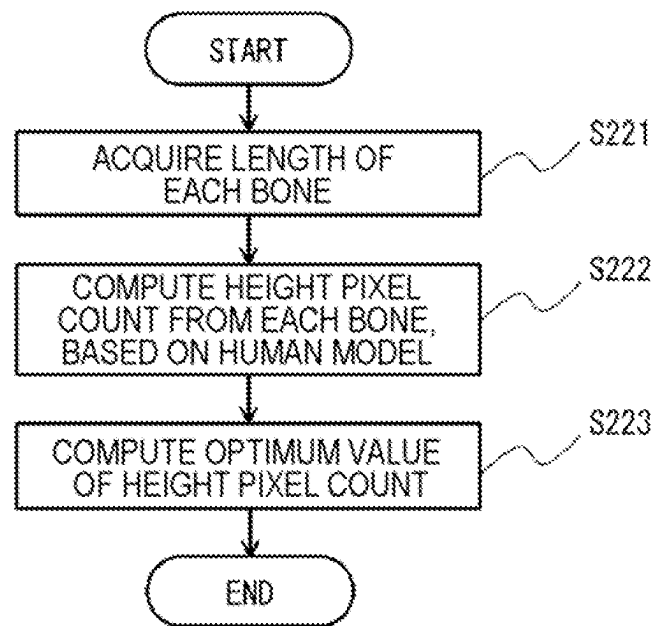


FIG. 22

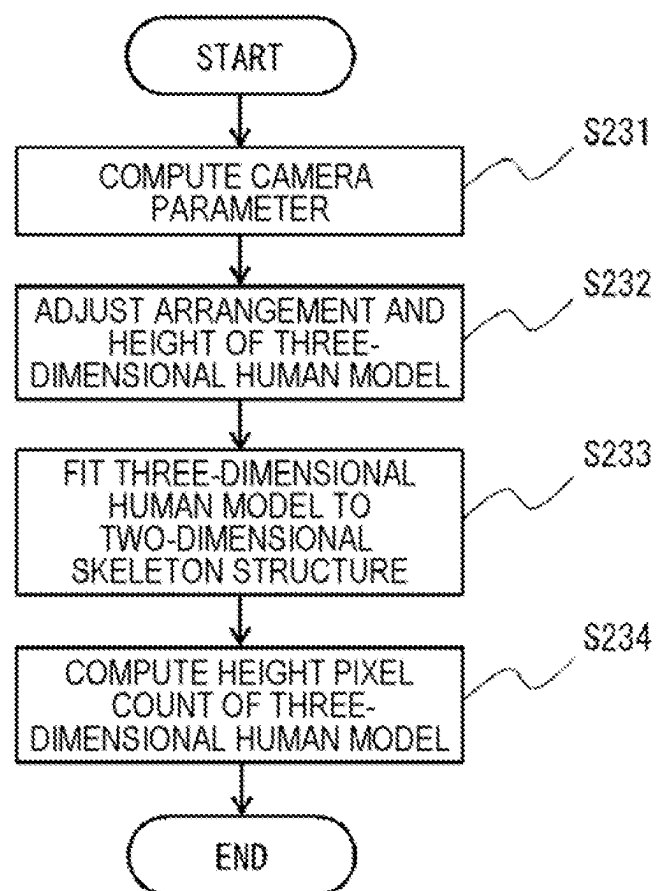


FIG. 23

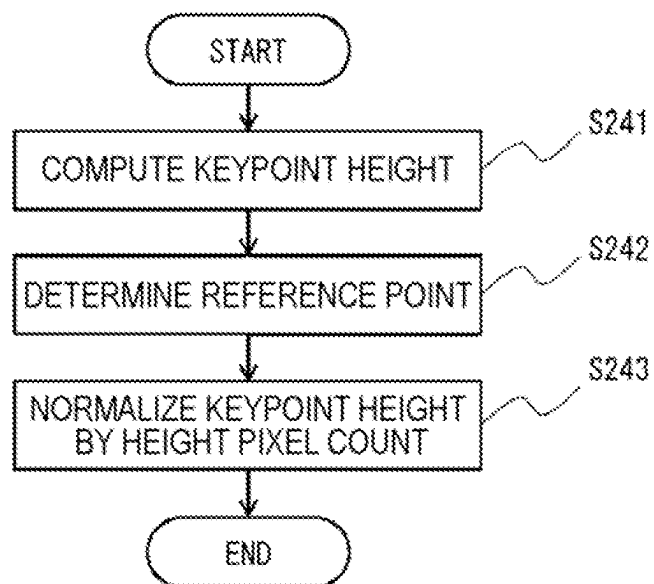


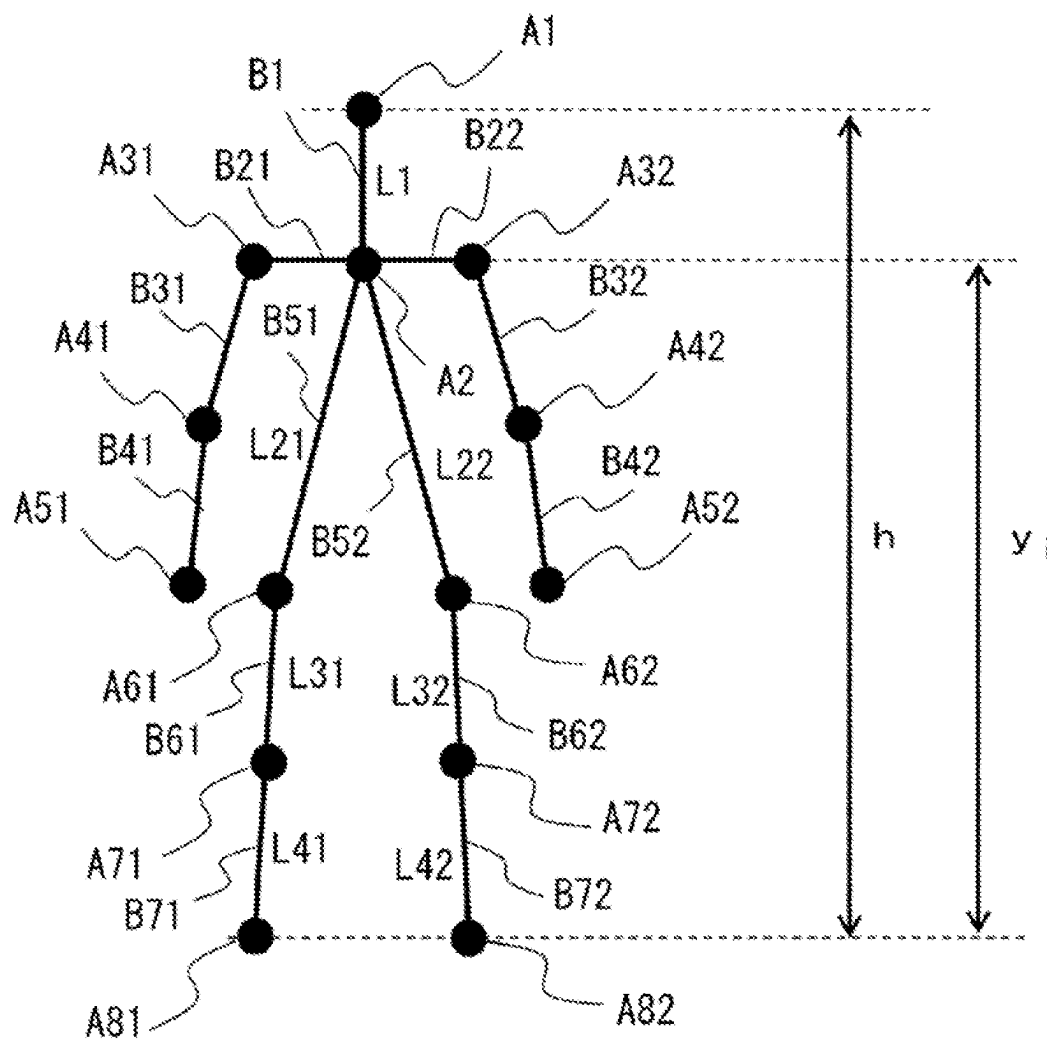
FIG. 24
300

FIG. 25

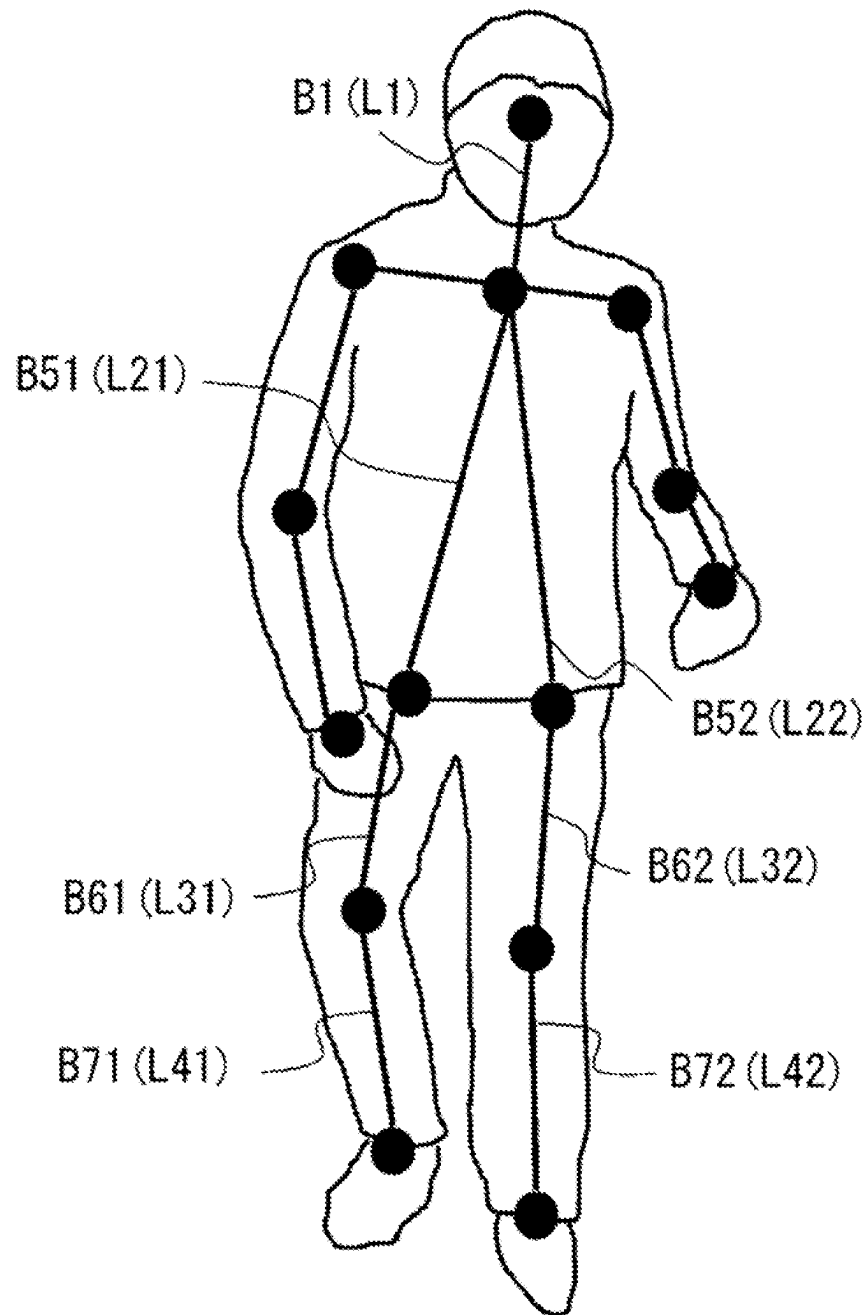


FIG. 26

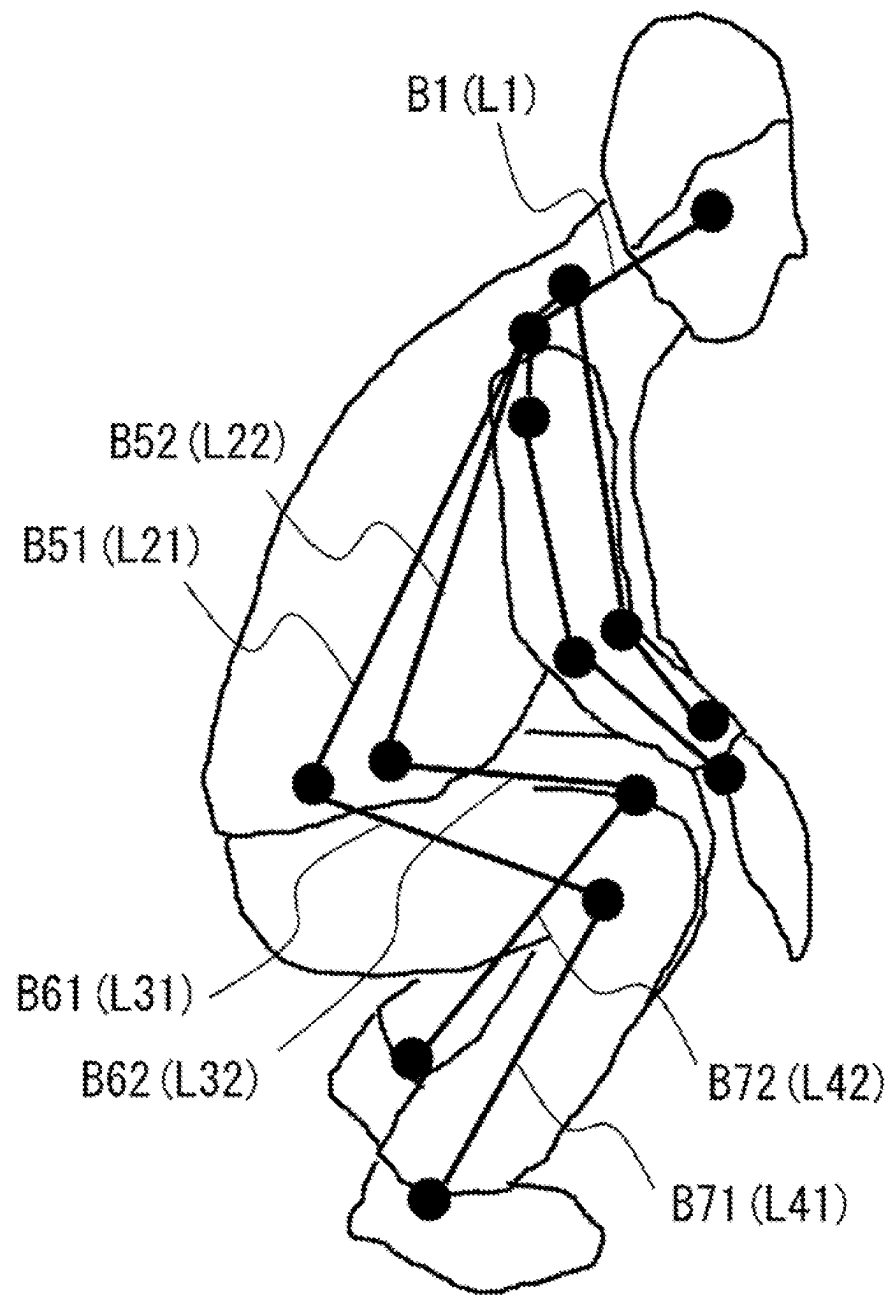


FIG. 27

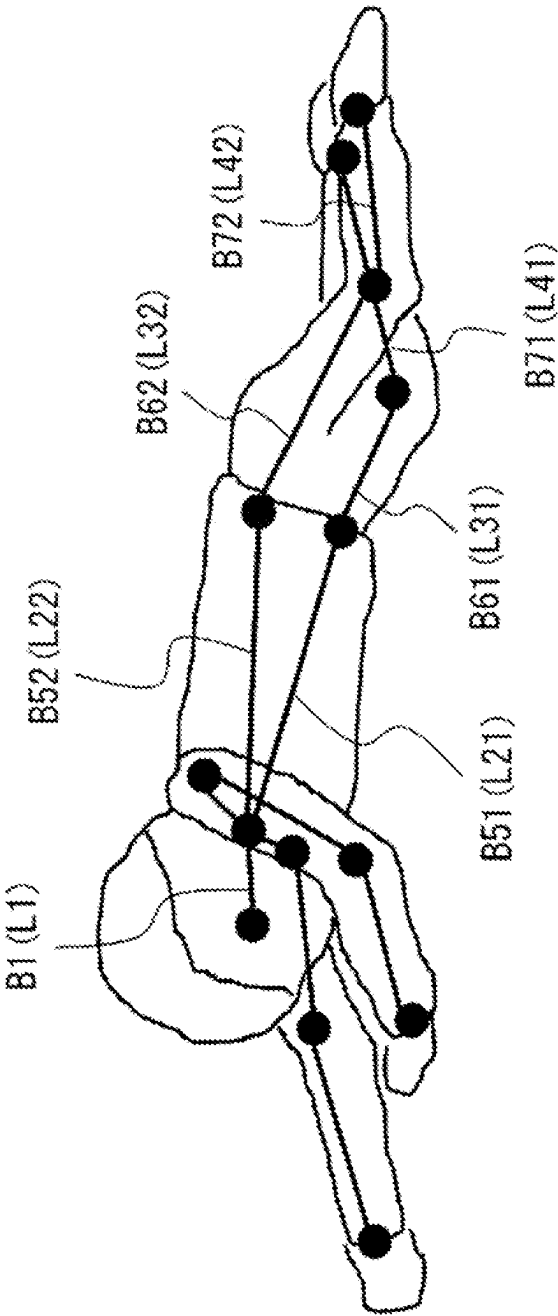


FIG. 28

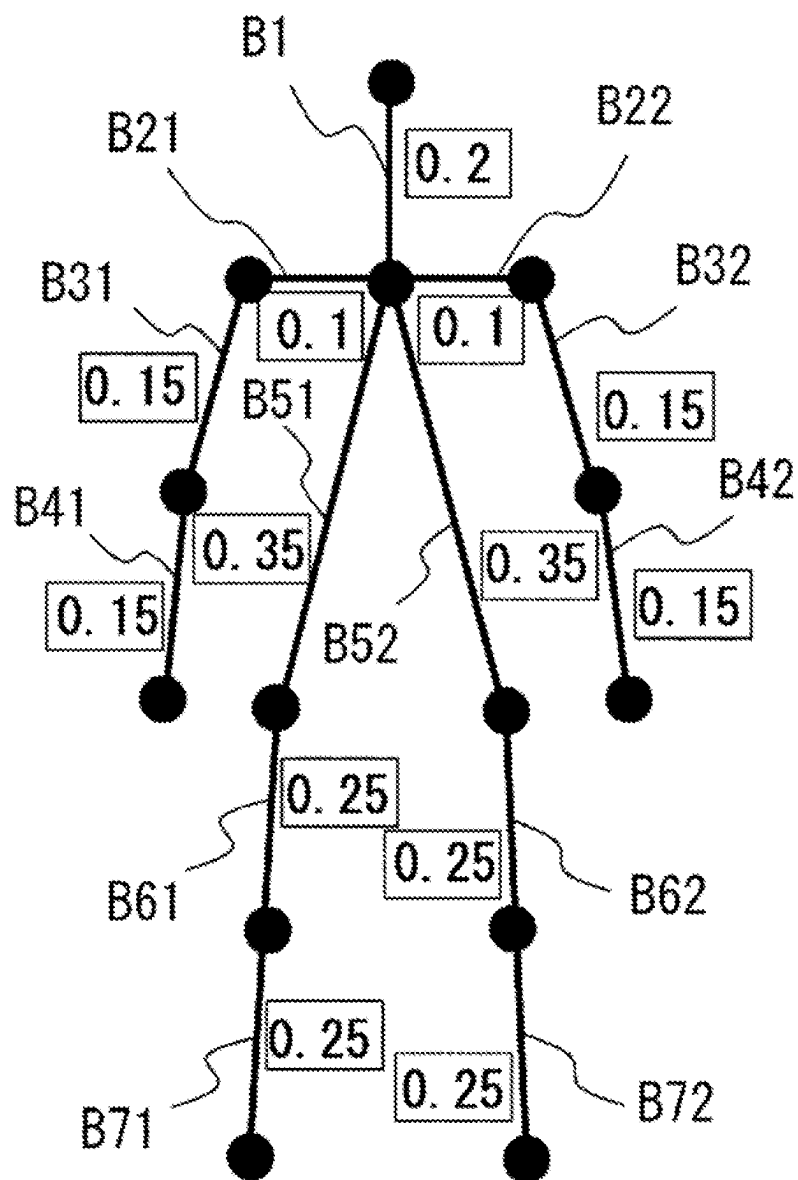
301

FIG. 29

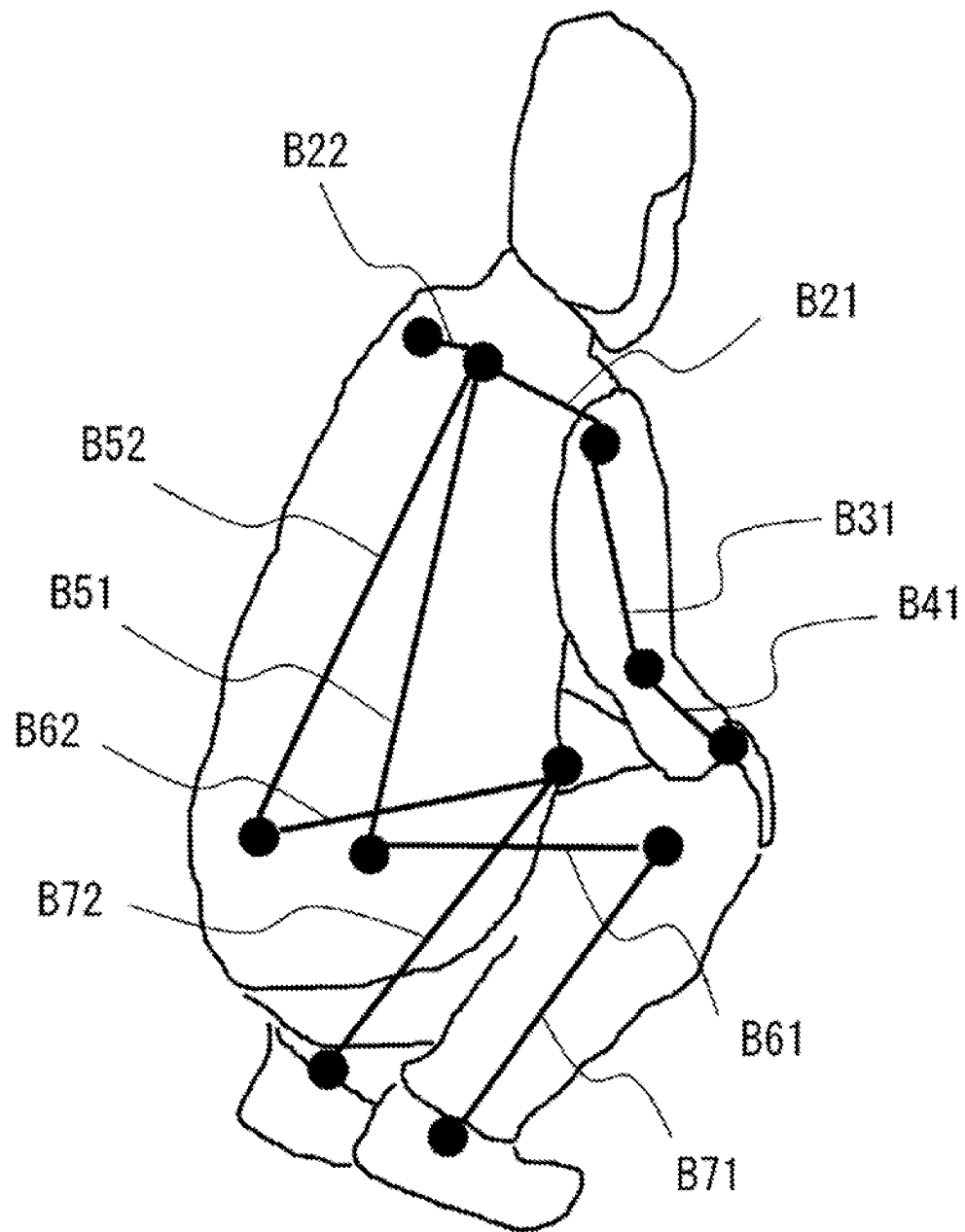


FIG. 30

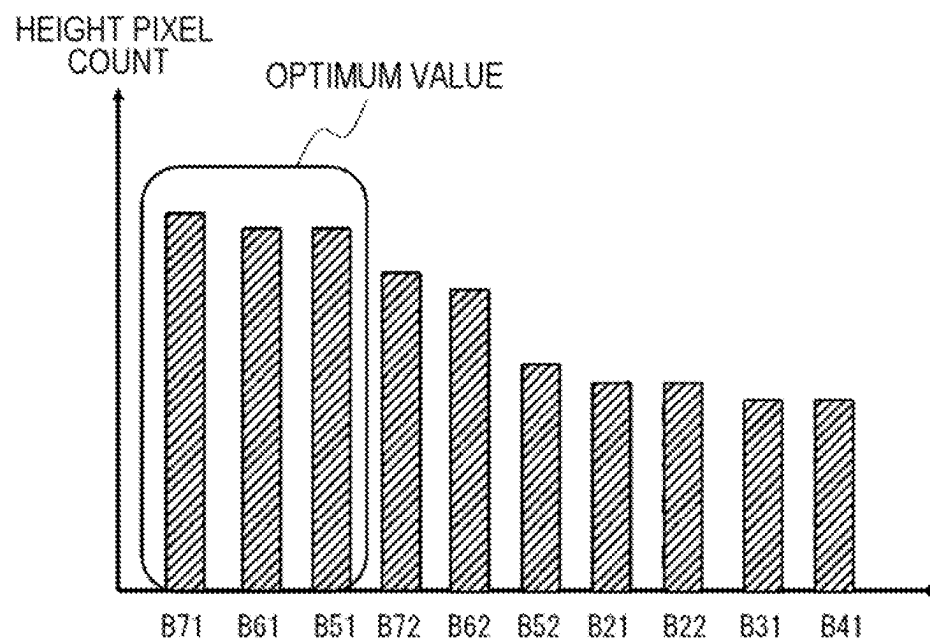
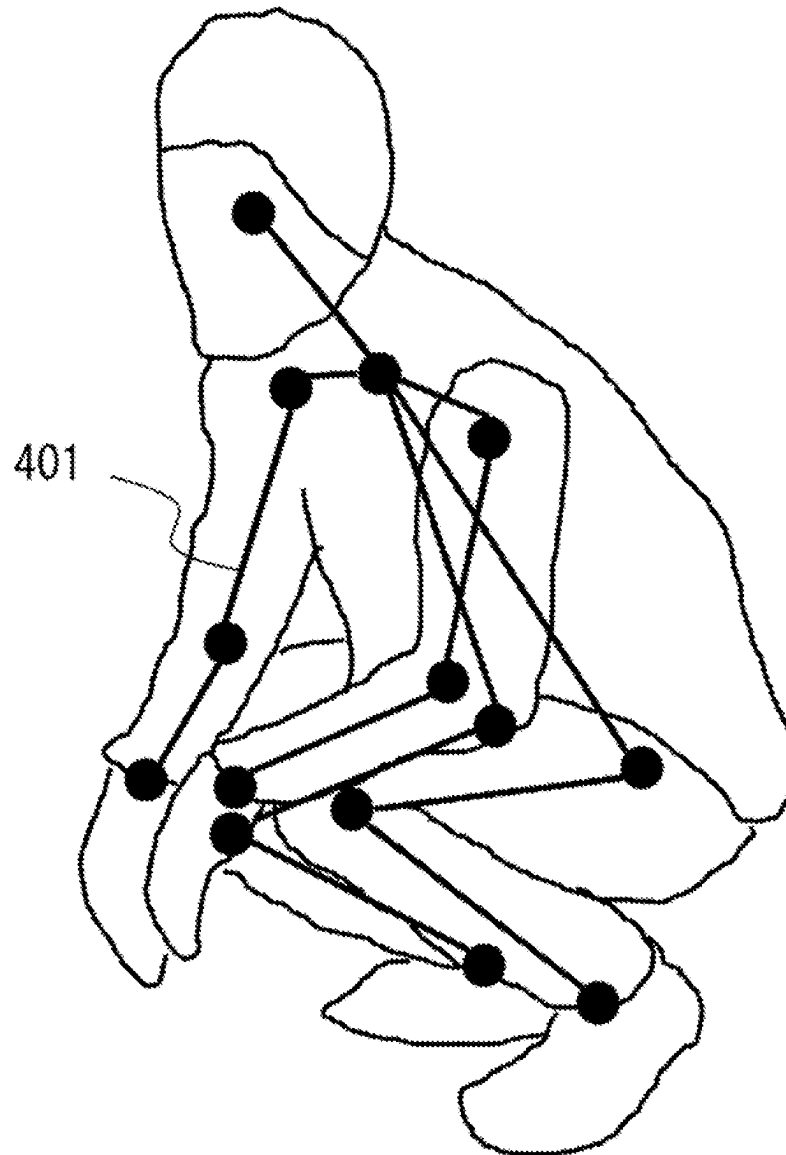


FIG. 31



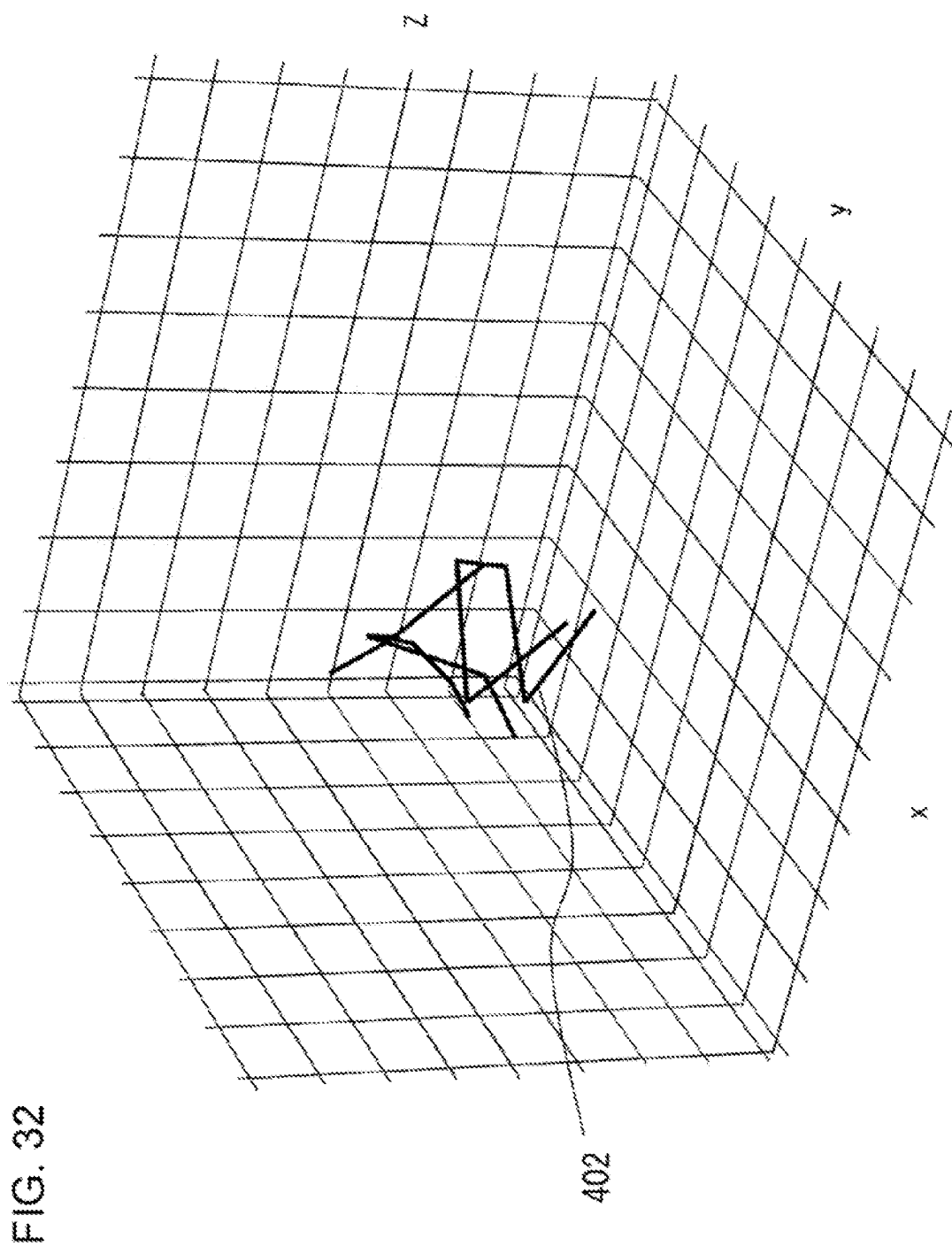


FIG. 33

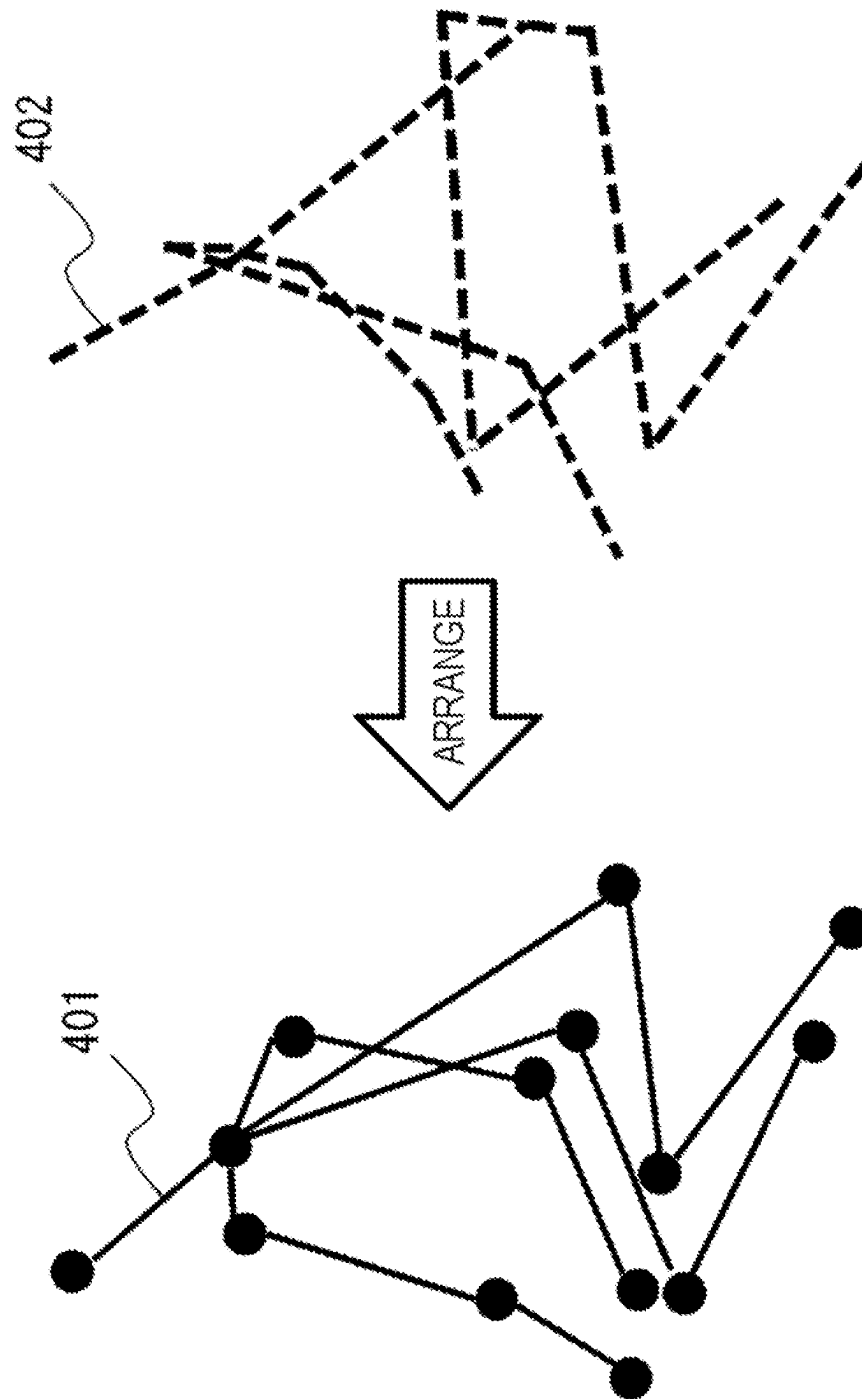


FIG. 34

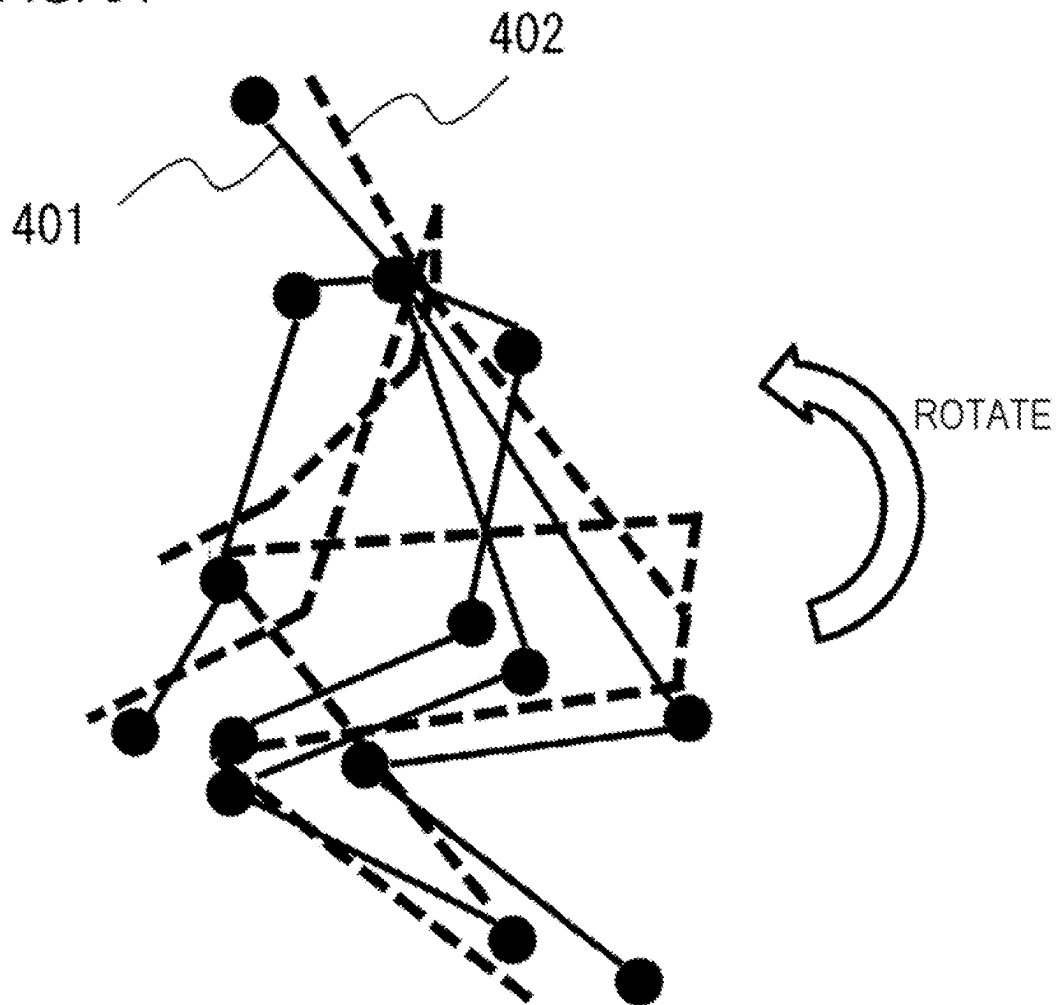


FIG. 35

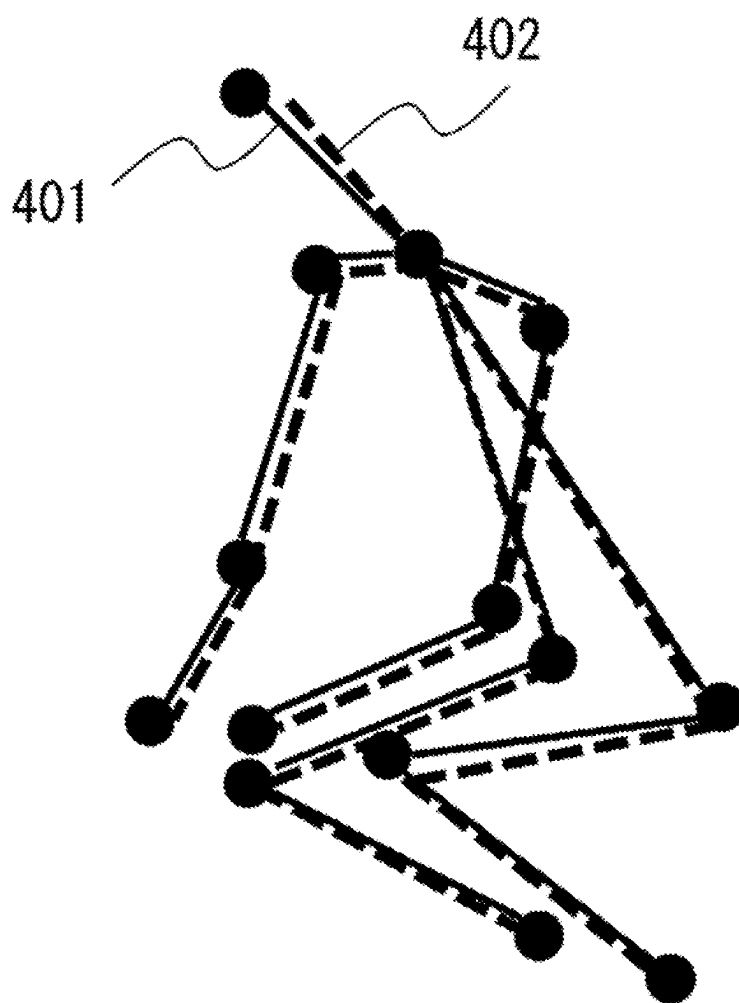


FIG. 36

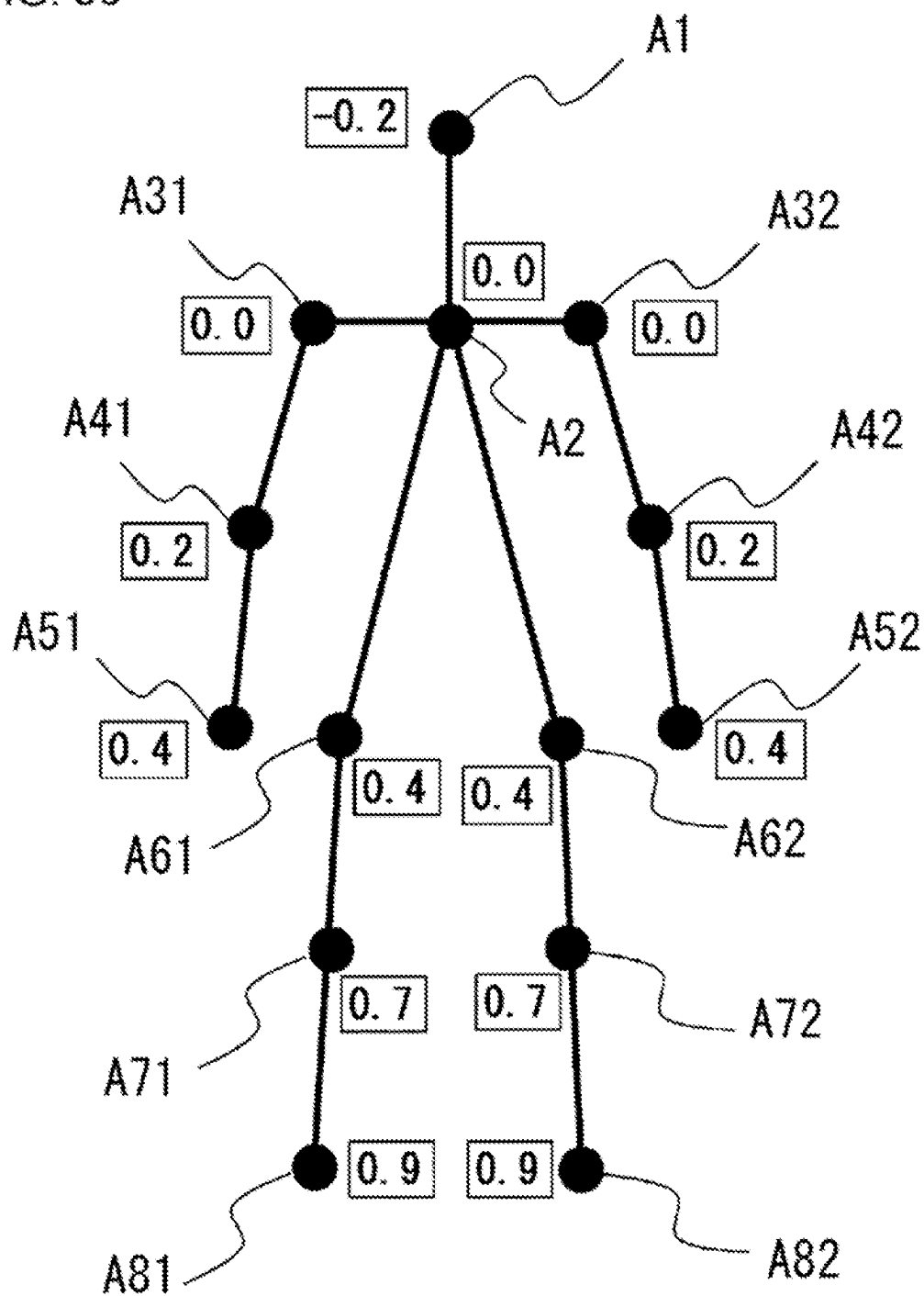


FIG. 37

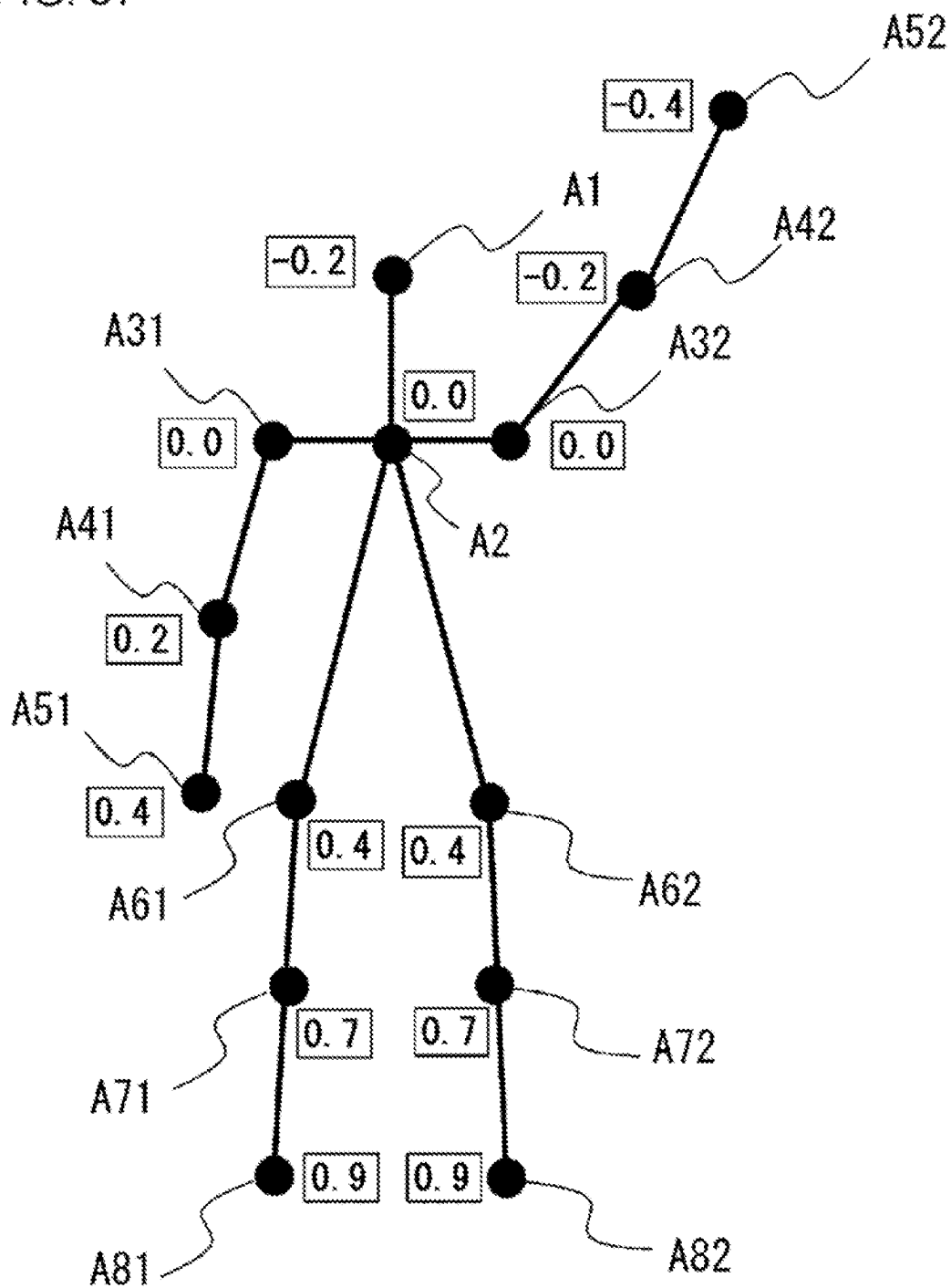


FIG. 38

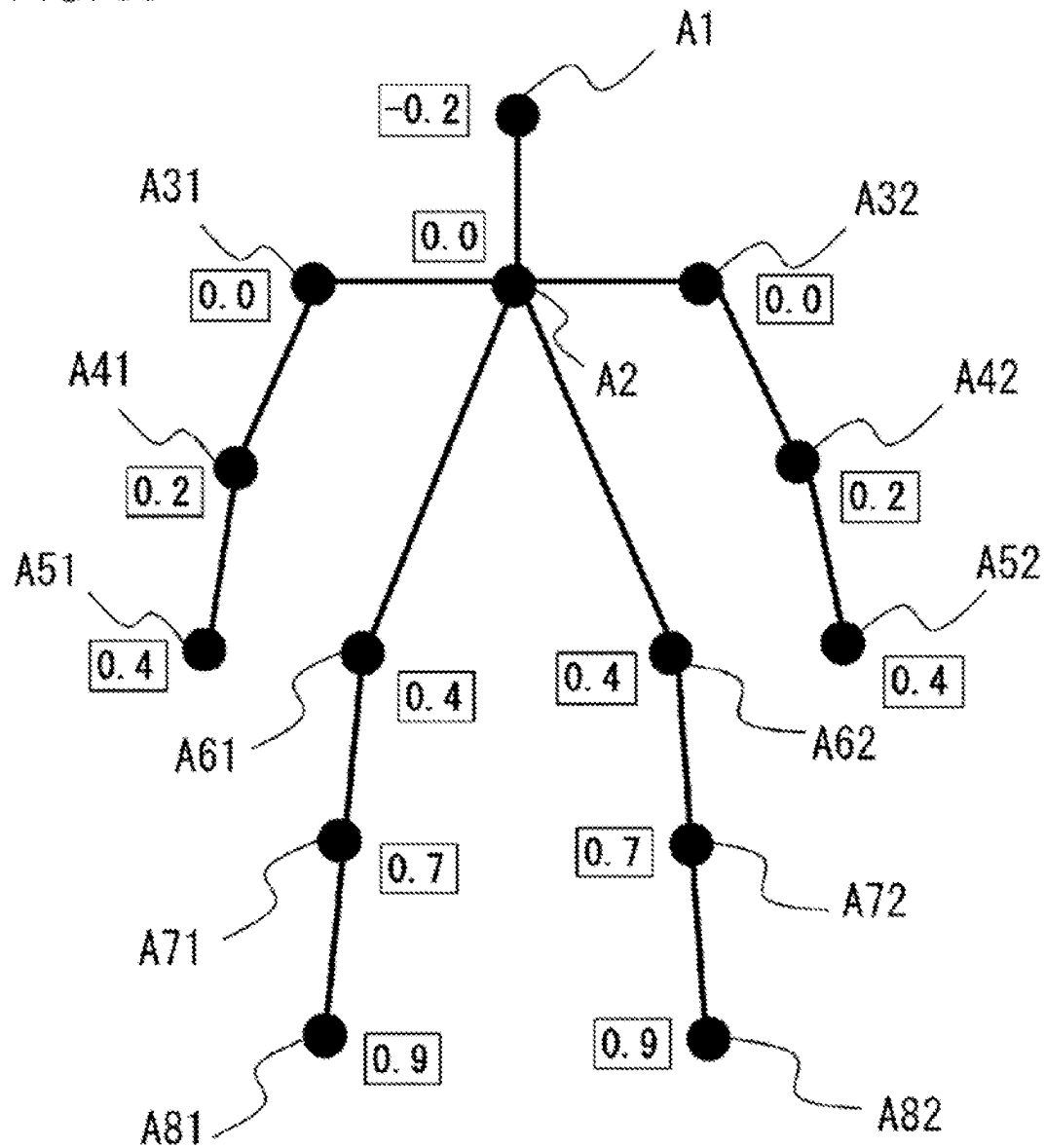


FIG. 39

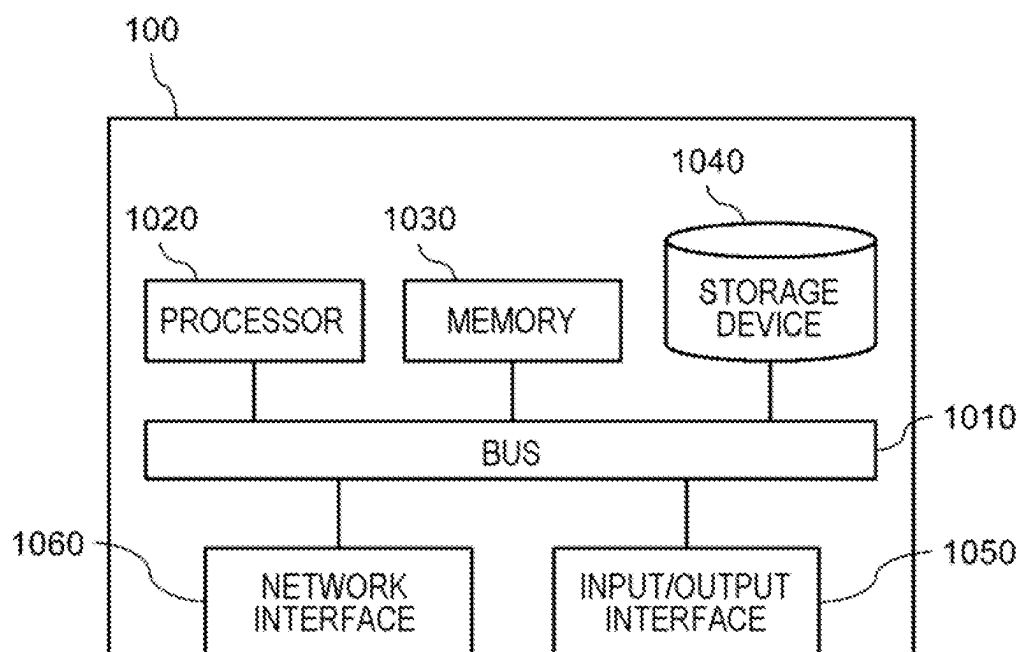


FIG. 40

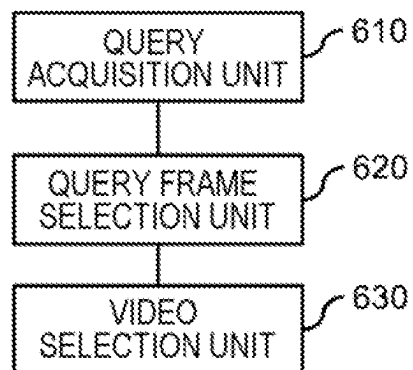
105

FIG. 41

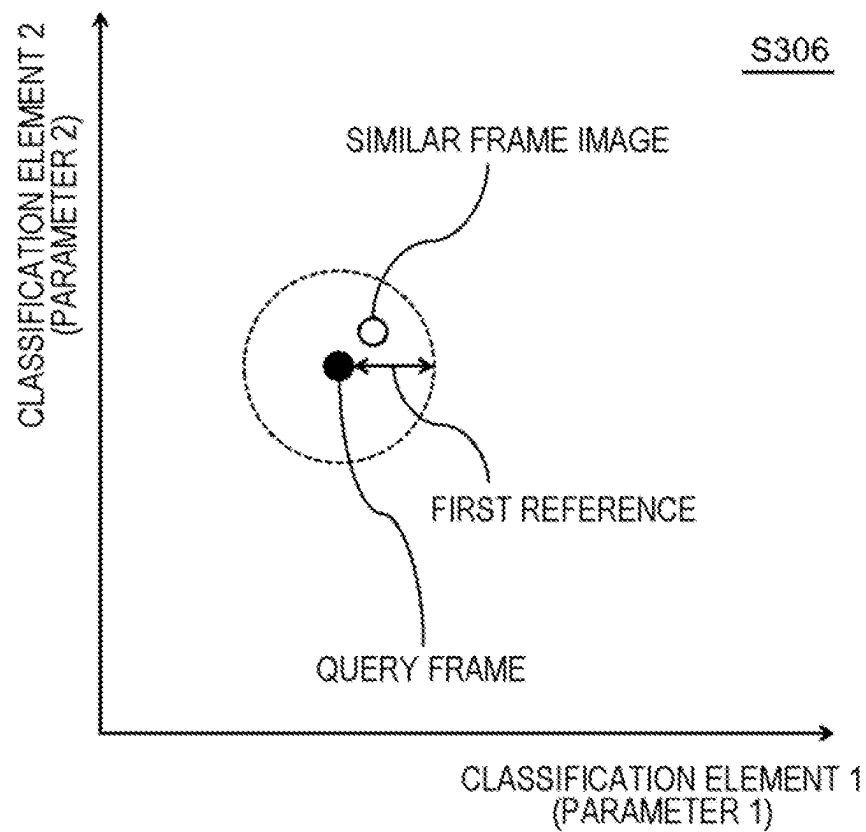


FIG. 42

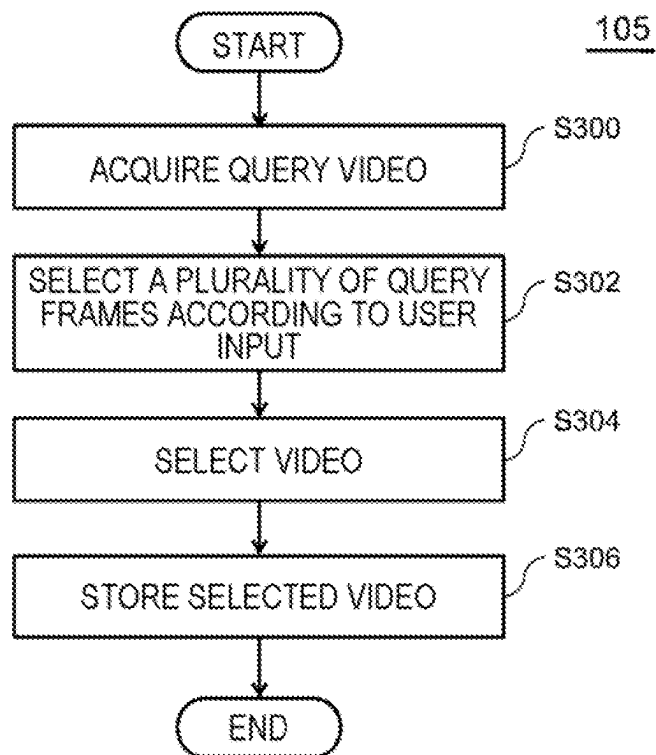


FIG. 43

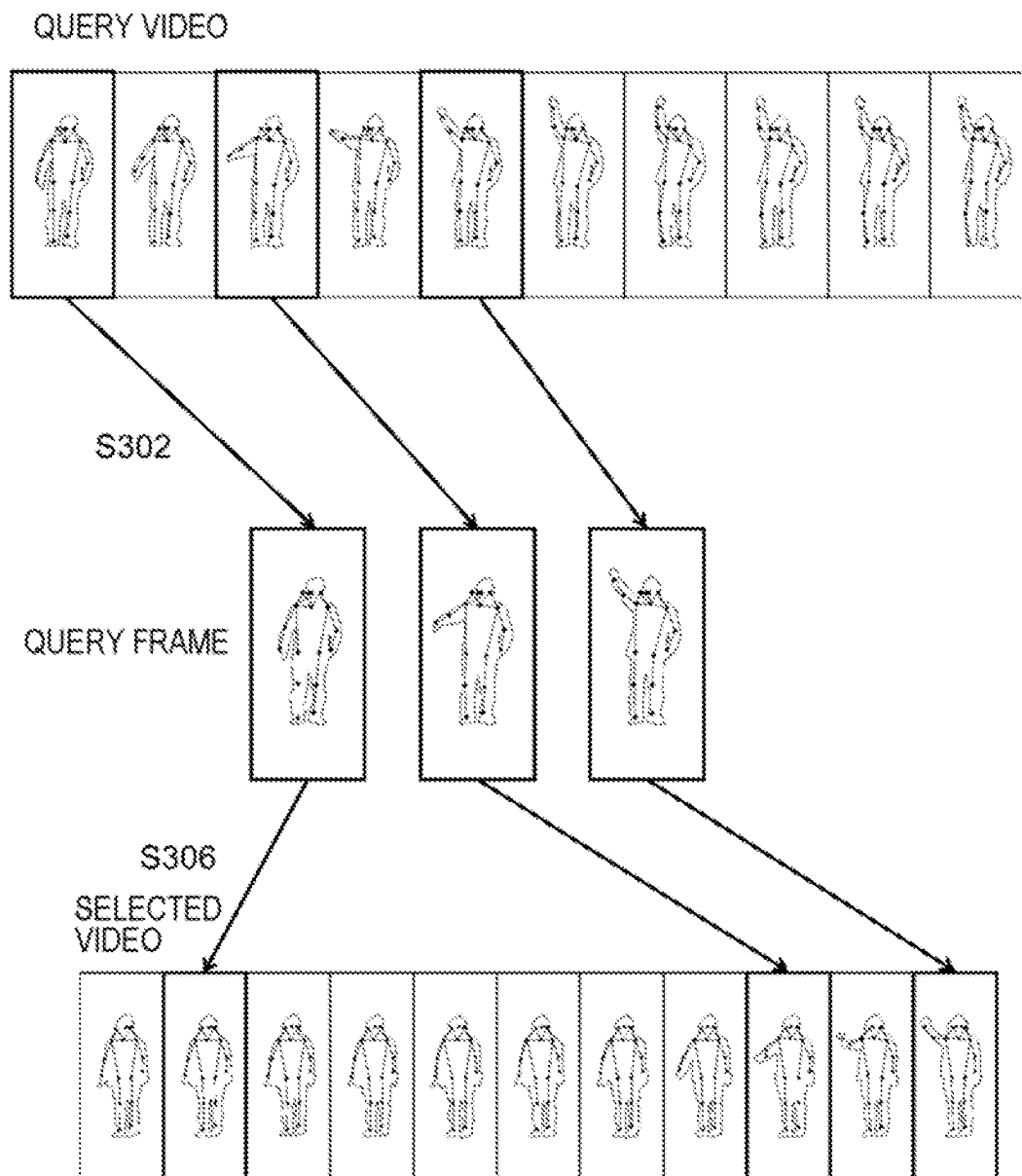


FIG. 44

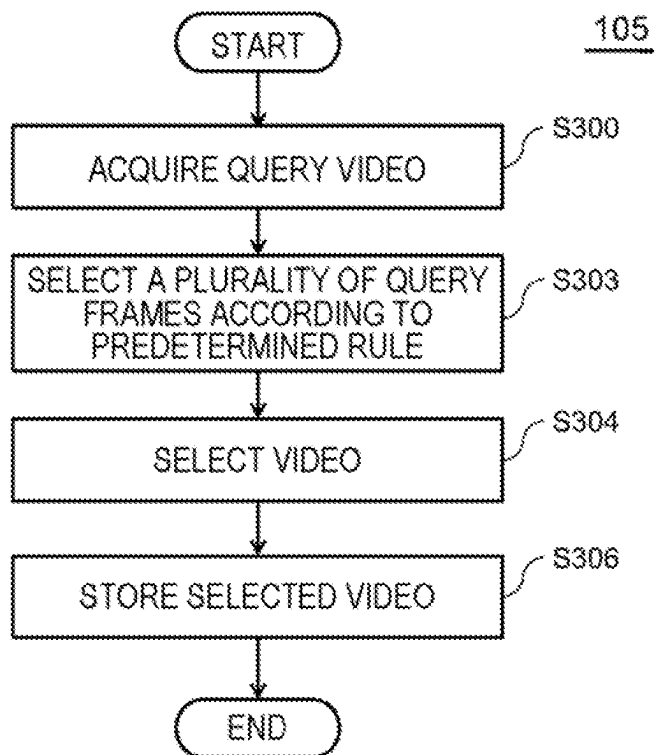


FIG. 45

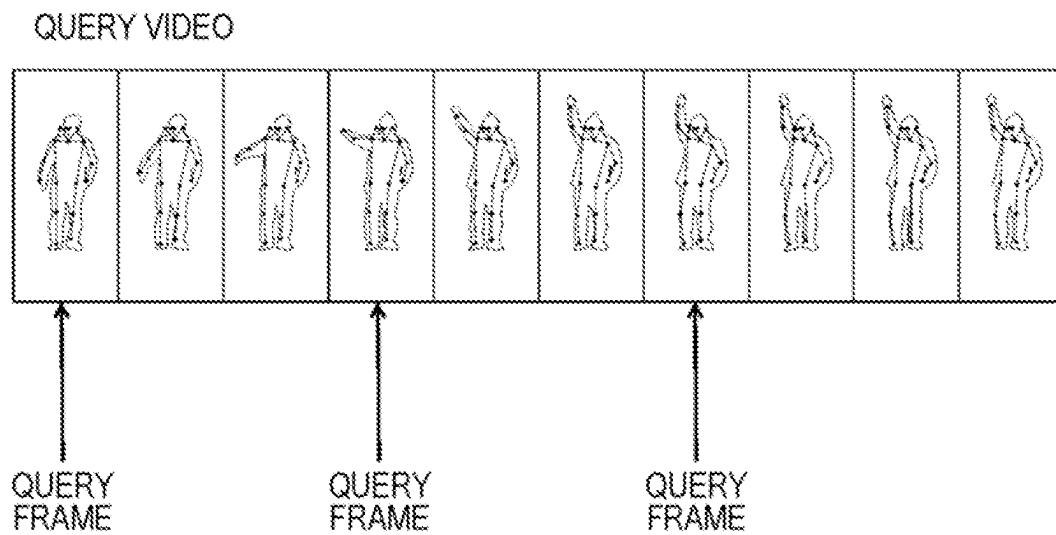


FIG. 46

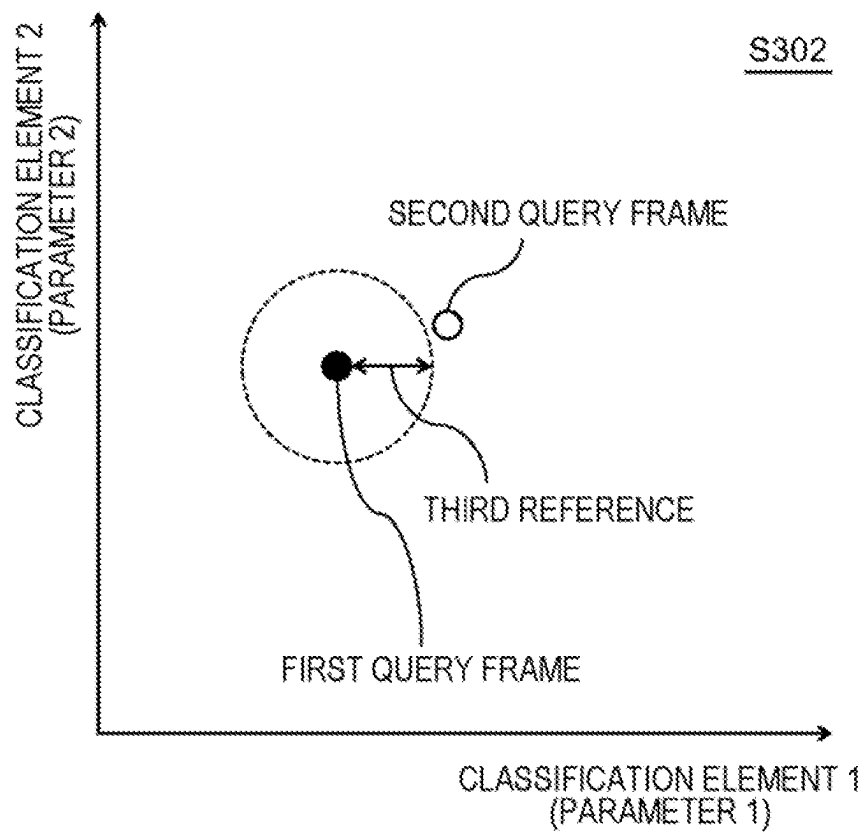
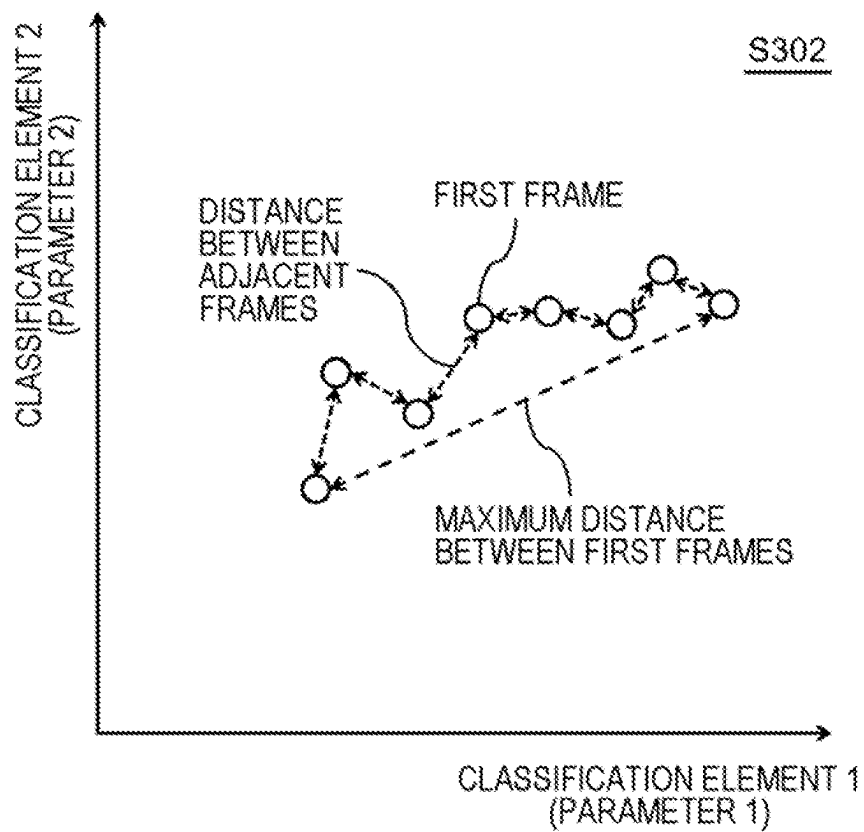


FIG. 47



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IMAGE SELECTION APPARATUS, IMAGE SELECTION METHOD, AND NON-TRANSITORY COMPUTER-READABLE MEDIUM

This application is a National Stage Entry of PCT/JP2020/019255 filed on May 14, 2020, the contents of all of which are incorporated herein by reference, in their entirety.

TECHNICAL FIELD

The present invention relates to an image selection apparatus, an image selection method, and a program.

BACKGROUND ART

In recent years, in a surveillance system and the like, a technique for detecting and searching for a state such as a pose and behavior of a person from an image of a surveillance camera is used. For example, Patent Documents 1 and 2 have been known as related techniques. Patent Document 1 discloses a technique for searching for a similar pose of a person, based on a key joint of a head, a hand, a foot, and the like of the person included in a depth video. Patent Document 2 discloses a technique for searching for a similar image by using pose information such as a tilt provided to an image, which is not related to a pose of a person. Note that, in addition, Non-Patent Document 1 has been known as a technique related to a skeleton estimation of a person.

Meanwhile, in recent years, it has been considered to use a video as a query and search for a video similar to the query. For example, Patent Document 3 describes that when a reference video to be a query is input, a similar video is searched for by using the number of faces of persons appearing, and a position, a size, and an orientation of a face of each person appearing.

RELATED DOCUMENT

Patent Document

Patent Document 1: Japanese Patent Application Publication (Translation of PCT Application) No. 2014-522035

Patent Document 2: Japanese Patent Application Publication No. 2006-260405

Patent Document 3: International Patent Publication No. WO2006/025272

Non-Patent Document

Non-Patent Document 1: Zhe Cao, Tomas Simon, Shih-En Wei, Yaser Sheikh, "Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields", The IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, P. 7291-7299

Disclosure of the Invention

SUMMARY OF THE INVENTION

Technical Problem

When a video is used as a query and a video similar to the query is selected, it is difficult to improve accuracy of selection. One of objects of the present invention is to

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improve accuracy of selection when a video is used as a query and a video similar to the query is selected.

Solution to Problem

The present invention provides an image selection apparatus including:

a query acquisition unit that acquires a query video including a plurality of first frame images;

a query frame selection unit that selects a plurality of query frames from among the plurality of first frame images; and

a video selection unit that selects a video by using the plurality of query frames, in which the video selection unit uses at least (1) and (2) below as a condition for selecting the video.

(1) A condition that a "similar frame image whose degree of similarity to the query frame satisfies a first reference is present" is satisfied for at least two of the query frames.

(2) When the query frames associated with each of the at least two similar frame images are arranged in a same order as that of the at least two similar frame images, the arrangement order coincides with an arrangement order of the at least two query frames in the query video.

The present invention provides an image selection method including, by a computer:

executing an acquisition step of acquiring a query video including a plurality of first frame images;

executing a query frame selection step of selecting a plurality of query frames from among the plurality of first frame images;

executing a video selection step of selecting a video by using the plurality of query frames; and,

in the video selection step, using at least (1) and (2) below as a condition for selecting the video.

(1) A condition that a "similar frame image whose degree of similarity to the query frame satisfies a first reference is present" is satisfied for at least two of the query frames.

(2) When the query frames associated with each of the at least two similar frame images are arranged in a same order as that of the at least two similar frame images, the arrangement order coincides with an arrangement order of the at least two query frames in the query video.

The present invention provides a program causing a computer to include:

an acquisition function of acquiring a query video including a plurality of first frame images;

a query frame selection function of selecting a plurality of query frames from among the plurality of first frame images; and

a video selection function of selecting a video by using the plurality of query frames, in which

the video selection function uses at least (1) and (2) below as a condition for selecting the video.

(1) A condition that a "similar frame image whose degree of similarity to the query frame satisfies a first reference is present" is satisfied for at least two of the query frames.

(2) When the query frames associated with each of the at least two similar frame images are arranged in a same order as that of the at least two similar frame images, the arrangement order coincides with an arrangement order of the at least two query frames in the query video.

According to the present invention, accuracy of selection improves when a video is used as a query and a video similar to the query is selected.

BRIEF DESCRIPTION OF THE DRAWINGS

The above-described object, the other objects, features, and advantages will become more apparent from suitable example embodiments described below and the following accompanying drawings.

FIG. 1 is a configuration diagram illustrating an outline of an image processing apparatus according to an example embodiment.

FIG. 2 is a configuration diagram illustrating a configuration of an image processing apparatus according to an example embodiment 1.

FIG. 3 is a flowchart illustrating an image processing method according to the example embodiment 1.

FIG. 4 is a flowchart illustrating a classification method according to the example embodiment 1.

FIG. 5 is a flowchart illustrating a search method according to the example embodiment 1.

FIG. 6 is a diagram illustrating a detection example of skeleton structures according to the example embodiment 1.

FIG. 7 is a diagram illustrating a human model according to the example embodiment 1.

FIG. 8 is a diagram illustrating a detection example of the skeleton structure according to the example embodiment 1.

FIG. 9 is a diagram illustrating a detection example of the skeleton structure according to the example embodiment 1.

FIG. 10 is a diagram illustrating a detection example of the skeleton structure according to the example embodiment 1.

FIG. 11 is a graph illustrating a specific example of the classification method according to the example embodiment 1.

FIG. 12 is a diagram illustrating a display example of a classification result according to the example embodiment 1.

FIG. 13 is a diagram for describing the search method according to the example embodiment 1.

FIG. 14 is a diagram for describing the search method according to the example embodiment 1.

FIG. 15 is a diagram for describing the search method according to the example embodiment 1.

FIG. 16 is a diagram for describing the search method according to the example embodiment 1.

FIG. 17 is a diagram illustrating a display example of a search result according to the example embodiment 1.

FIG. 18 is a configuration diagram illustrating a configuration of an image processing apparatus according to an example embodiment 2.

FIG. 19 is a flowchart illustrating an image processing method according to the example embodiment 2.

FIG. 20 is a flowchart illustrating a specific example 1 of a height pixel count computation method according to the example embodiment 2.

FIG. 21 is a flowchart illustrating a specific example 2 of the height pixel count computation method according to the example embodiment 2.

FIG. 22 is a flowchart illustrating the specific example 3 of the height pixel count computation method according to the example embodiment 2.

FIG. 23 is a flowchart illustrating a normalization method according to the example embodiment 2.

FIG. 24 is a diagram illustrating a human model according to the example embodiment 2.

FIG. 25 is a diagram illustrating a detection example of a skeleton structure according to the example embodiment 2.

FIG. 26 is a diagram illustrating a detection example of a skeleton structure according to the example embodiment 2.

FIG. 27 is a diagram illustrating a detection example of a skeleton structure according to the example embodiment 2.

FIG. 28 is a diagram illustrating a human model according to the example embodiment 2.

FIG. 29 is a diagram illustrating a detection example of a skeleton structure according to the example embodiment 2.

FIG. 30 is a histogram for describing the height pixel count computation method according to the example embodiment 2.

FIG. 31 is a diagram illustrating a detection example of a skeleton structure according to the example embodiment 2.

FIG. 32 is a diagram illustrating a three-dimensional human model according to the example embodiment 2.

FIG. 33 is a diagram for describing the height pixel count computation method according to the example embodiment 2.

FIG. 34 is a diagram for describing the height pixel count computation method according to the example embodiment 2.

FIG. 35 is a diagram for describing the height pixel count computation method according to the example embodiment 2.

FIG. 36 is a diagram for describing the normalization method according to the example embodiment 2.

FIG. 37 is a diagram for describing the normalization method according to the example embodiment 2.

FIG. 38 is a diagram for describing the normalization method according to the example embodiment 2.

FIG. 39 is a diagram illustrating a hardware configuration example of the image processing apparatus.

FIG. 40 is a diagram illustrating a functional configuration of a search unit according to a search method 6.

FIG. 41 is a diagram for describing one example of a method for selecting a similar frame by a video selection unit.

FIG. 42 is a flowchart illustrating one example of processing performed by the search unit.

FIG. 43 is a diagram for describing the processing illustrated in FIG. 42.

FIG. 44 is a flowchart illustrating a modification example of FIG. 42.

FIG. 45 is a diagram for describing a first example of a selection rule of a query frame.

FIG. 46 is a diagram for describing a second example of the selection rule of a query frame.

FIG. 47 is a diagram for describing a method for setting a third reference.

DESCRIPTION OF EMBODIMENTS

Hereinafter, example embodiments of the present invention will be described with reference to the drawings. Note that, in all of the drawings, a similar component has a similar reference sign, and description thereof will not be appropriately repeated.

Consideration for Example Embodiment

In recent years, an image recognition technique using machine learning such as deep learning is applied to various systems. For example, application to a surveillance system

for performing surveillance by an image of a surveillance camera has been advanced. By using machine learning for the surveillance system, a state such as a pose and behavior of a person is becoming recognizable from an image to some extent.

However, in such a related technique, a state of a person desired by a user may not be necessarily recognizable on demand. For example, there is a case where a state of a person desired to be searched for and recognized by a user can be determined in advance, or there is a case where a determination cannot be specifically made as in an unknown state. Thus, in some cases, a state of a person desired to be searched for by a user cannot be specified in detail. Further, a search or the like cannot be performed when a part of a body of a person is hidden. In the related technique, a state of a person can be searched for only from a specific search condition, and thus it is difficult to flexibly search for and classify a desired state of a person.

Thus, the inventors have considered a method using a skeleton estimation technique such as Non-Patent Document 1 and the like in order to recognize a state of a person desired by a user from an image on demand. Similarly to Open Pose disclosed in Non-Patent Document 1, and the like, in the related skeleton estimation technique, a skeleton of a person is estimated by learning image data in which correct answers in various patterns are set. In the following example embodiments, a state of a person can be flexibly recognized by using such a skeleton estimation technique.

Note that, a skeleton structure estimated by the skeleton estimation technique such as Open Pose is formed of a “keypoint” being a characteristic point such as a joint and a “bone (bone link)” indicating a link between keypoints. Thus, in the following example embodiments, the words “keypoint” and “bone” will be used to describe a skeleton structure, and “keypoint” is associated with a “joint” of a person and “bone” is associated with a “bone” of a person unless otherwise specified. Then, a position of the “keypoint” is one example of joint information.

Overview of Example Embodiment

FIG. 1 illustrates an outline of an image processing apparatus 10 according to an example embodiment. As illustrated in FIG. 1, the image processing apparatus 10 includes a skeleton detection unit 11, a feature value computation unit 12, and a recognition unit 13. The skeleton detection unit 11 detects two-dimensional skeleton structures of a plurality of persons, based on a two-dimensional image acquired from a camera and the like. The feature value computation unit 12 computes feature values of the plurality of two-dimensional skeleton structures detected by the skeleton detection unit 11. The recognition unit 13 performs recognition processing of a state of the plurality of persons, based on a degree of similarity between the plurality of feature values computed by the feature value computation unit 12. The recognition processing is classification processing, search processing (selection processing), and the like of a state of a person. Thus, the image processing apparatus 10 also functions as an image selection apparatus.

In this way, in the example embodiment, a two-dimensional skeleton structure of a person is detected from a two-dimensional image, and the recognition processing such as classification and search of a state of a person is performed based on a feature value computed from the two-

dimensional skeleton structure, and thus a desired state of a person can be flexibly recognized.

Example Embodiment 1

An example embodiment 1 will be described below with reference to the drawings. FIG. 2 illustrates a configuration of an image processing apparatus 100 according to the present example embodiment. The image processing apparatus 100 constitutes an image processing system 1, together with a camera 200 and a database (DB) 110. The image processing system 1 including the image processing apparatus 100 is a system for classifying and searching for a state such as a pose and behavior of a person, based on a skeleton structure of the person estimated from an image. Note that, the image processing apparatus 100 also functions as an image selection apparatus.

The camera 200 is an image capturing unit, such as a surveillance camera, that generates a two-dimensional image. The camera 200 is installed at a predetermined place, and captures an image of a person and the like in the imaging area from the installed place. The camera 200 may be directly connected to the image processing apparatus 100 in such a way as to be able to output a captured image (video) to the image processing apparatus 100, or may be connected to the image processing apparatus 100 via a network and the like. Note that, the camera 200 may be provided inside the image processing apparatus 100.

The database 110 is a database that stores information (data) needed for processing of the image processing apparatus 100, a processing result, and the like. The database 110 stores an image acquired by an image acquisition unit 101, a detection result of a skeleton structure detection unit 102, data for machine learning, a feature value computed by a feature value computation unit 103, a classification result of a classification unit 104, a search result of a search unit 105, and the like. The database 110 is directly connected to the image processing apparatus 100 in such a way as to be able to input and output data as necessary, or is connected to the image processing apparatus 100 via a network and the like. Note that, the database 110 may be provided inside the image processing apparatus 100 as a non-volatile memory such as a flash memory, a hard disk apparatus, and the like.

As illustrated in FIG. 2, the image processing apparatus 100 includes the image acquisition unit 101, the skeleton structure detection unit 102, the feature value computation unit 103, the classification unit 104, the search unit 105, an input unit 106, and a display unit 107. Note that, a configuration of each unit (block) is one example, and another unit may be used for a configuration as long as a method (operation) described below can be achieved. Further, the image processing apparatus 100 is achieved by a computer apparatus, such as a personal computer and a server, that executes a program, for example, but may be achieved by one apparatus or may be achieved by a plurality of apparatuses on a network. For example, the input unit 106, the display unit 107, and the like may be an external apparatus. Further, both of the classification unit 104 and the search unit 105 may be provided, or only one of them may be provided. Both or one of the classification unit 104 and the search unit 105 is a recognition unit that performs the recognition processing of a state of a person.

The image acquisition unit 101 acquires a two-dimensional image including a person captured by the camera 200. The image acquisition unit 101 acquires an image (video including a plurality of images) including a person captured by the camera 200 in a predetermined surveillance period,

for example. Note that, instead of acquisition from the camera **200**, an image including a person being prepared in advance may be acquired from the database **110** and the like.

The skeleton structure detection unit **102** detects a two-dimensional skeleton structure of the person in the acquired two-dimensional image, based on the image. The skeleton structure detection unit **102** detects a skeleton structure for all persons recognized in the acquired image. The skeleton structure detection unit **102** detects a skeleton structure of a recognized person, based on a feature such as a joint of the person, by using a skeleton estimation technique using machine learning. The skeleton structure detection unit **102** uses a skeleton estimation technique such as Open Pose in Non-Patent Document 1, for example.

The feature value computation unit **103** computes a feature value of the detected two-dimensional skeleton structure, and stores, in the database **110**, the computed feature value in association with the image to be processed. The feature value of the skeleton structure indicates a feature of a skeleton of the person, and is an element for classifying and searching for a state of the person, based on the skeleton of the person. This feature value normally includes a plurality of parameters (for example, a classification element described below). Then, the feature value may be a feature value of the entire skeleton structure, may be a feature value of a part of the skeleton structure, or may include a plurality of feature values as in each portion of the skeleton structure. A method for computing a feature value may be any method such as machine learning and normalization, and a minimum value and a maximum value may be acquired as normalization. As one example, the feature value is a feature value acquired by performing machine learning on the skeleton structure, a size of the skeleton structure from a head to a foot on an image, and the like. The size of the skeleton structure is a height in an up-down direction, an area, and the like of a skeleton region including the skeleton structure on an image. The up-down direction (a height direction or a vertical direction) is a direction (Y-axis direction) of up and down in an image, and is, for example, a direction perpendicular to the ground (reference surface). Further, a left-right direction (a horizontal direction) is a direction (X-axis direction) of left and right in an image, and is, for example, a direction parallel to the ground.

Note that, in order to perform classification and a search desired by a user, a feature value having robustness with respect to classification and search processing is preferably used. For example, when a user desires classification and a search that do not depend on an orientation and a body shape of a person, a feature value that is robust with respect to the orientation and the body shape of the person may be used. A feature value that does not depend on an orientation and a body shape of a person can be acquired by learning skeletons of persons facing in various directions with the same pose and skeletons of persons having various body shapes with the same pose, and extracting a feature only in the up-down direction of a skeleton.

The classification unit **104** classifies a plurality of skeleton structures stored in the database **110**, based on a degree of similarity between feature values of the skeleton structures (performs clustering). It can also be said that, as the recognition processing of a state of a person, the classification unit **104** classifies states of a plurality of persons, based on feature values of the skeleton structures. The degree of similarity is a distance between the feature values of the skeleton structures. The classification unit **104** may perform classification by a degree of similarity between feature values of the entire skeleton structures, may perform clas-

sification by a degree of similarity between feature values of a part of the skeleton structures, and may perform classification by a degree of similarity between feature values of a first portion (for example, both hands) and a second portion (for example, both feet) of the skeleton structures. Note that, a pose of a person may be classified based on a feature value of a skeleton structure of the person in each image, and behavior of a person may be classified based on a change in a feature value of a skeleton structure of the person in a plurality of images successive in time series. In other words, the classification unit **104** may classify a state of a person including a pose and behavior of the person, based on a feature value of a skeleton structure. For example, the classification unit **104** sets, as subjects to be classified, a plurality of skeleton structures in a plurality of images captured in a predetermined surveillance period. The classification unit **104** acquires a degree of similarity between feature values of the subjects to be classified, and performs classification in such a way that skeleton structures having a high degree of similarity are in the same cluster (group with a similar pose). Note that, similarly to a search, a user may be able to specify a classification condition. The classification unit **104** stores a classification result of the skeleton structure in the database **110**, and also displays the classification result on the display unit **107**.

The search unit **105** searches for a skeleton structure having a high degree of similarity to a feature value of a search query (query state) from among the plurality of skeleton structures stored in the database **110**. It can also be said that, as the recognition processing of a state of a person, the search unit **105** searches for a state of a person that corresponds to a search condition (query state) from among states of a plurality of persons, based on feature values of the skeleton structures. Similarly to classification, the degree of similarity is a distance between the feature values of the skeleton structures. The search unit **105** may perform a search by a degree of similarity between feature values of the entire skeleton structures, may perform a search by a degree of similarity between feature values of a part of the skeleton structures, and may perform a search by a degree of similarity between feature values of a first portion (for example, both hands) and a second portion (for example, both feet) of the skeleton structures. Note that, a pose of a person may be searched based on a feature value of a skeleton structure of the person in each image, and behavior of a person may be searched based on a change in a feature value of a skeleton structure of the person in a plurality of images successive in time series. In other words, the search unit **105** can search for a state of a person including a pose and behavior of the person, based on a feature value of a skeleton structure. For example, similarly to subjects to be classified, the search unit **105** sets, as subjects to be searched, feature values of a plurality of skeleton structures in a plurality of images captured in a predetermined surveillance period. Further, a skeleton structure (pose) specified by a user from among classification results displayed on the classification unit **104** is set as a search query (search key). Note that, without limitation to a classification result, a search query may be selected from among a plurality of skeleton structures that are not classified, or a user may input a skeleton structure to be a search query. The search unit **105** searches for a feature value having a high degree of similarity to a feature value of a skeleton structure being a search query from among feature values being subjects to be searched. The search unit **105** stores a search result of the feature value in the database **110**, and also displays the search result on the display unit **107**.

The input unit **106** is an input interface that acquires information input by a user who operates the image processing apparatus **100**. For example, the user is a surveillant who watches a person in a suspicious state from an image of a surveillance camera. The input unit **106** is, for example, a graphical user interface (GUI), and receives an input of information according to an operation of the user from an input apparatus such as a keyboard, a mouse, and a touch panel. For example, the input unit **106** receives, as a search query, a skeleton structure of a person specified from among the skeleton structures (poses) classified by the classification unit **104**.

The display unit **107** is a display unit that displays a result of an operation (processing) of the image processing apparatus **100**, and the like, and is, for example, a display apparatus such as a liquid crystal display and an organic electro luminescence (EL) display. The display unit **107** displays, on the GUI, a classification result of the classification unit **104** and a search result of the search unit **105** according to a degree of similarity and the like.

FIG. 39 is a diagram illustrating a hardware configuration example of the image processing apparatus **100**. The image processing apparatus **100** includes a bus **1010**, a processor **1020**, a memory **1030**, a storage device **1040**, an input/output interface **1050**, and a network interface **1060**.

The bus **1010** is a data transmission path for allowing the processor **1020**, the memory **1030**, the storage device **1040**, the input/output interface **1050**, and the network interface **1060** to transmit and receive data with one another. However, a method of connecting the processor **1020** and the like to each other is not limited to bus connection.

The processor **1020** is a processor achieved by a central processing unit (CPU), a graphics processing unit (GPU), and the like.

The memory **1030** is a main storage achieved by a random access memory (RAM) and the like.

The storage device **1040** is an auxiliary storage achieved by a hard disk drive (HDD), a solid state drive (SSD), a memory card, a read only memory (ROM), or the like. The storage device **1040** stores a program module that achieves each function (for example, the image acquisition unit **101**, the skeleton structure detection unit **102**, the feature value computation unit **103**, the classification unit **104**, the search unit **105**, and the input unit **106**) of the image processing apparatus **100**. The processor **1020** reads each program module onto the memory **1030** and executes the program module, and each function associated with the program module is achieved. Further, the storage device **1040** may also function as the database **110**.

The input/output interface **1050** is an interface for connecting the image processing apparatus **100** and various types of input/output equipment. When the database **110** is located outside the image processing apparatus **100**, the image processing apparatus **100** may be connected to the database **110** via the input/output interface **1050**.

The network interface **1060** is an interface for connecting the image processing apparatus **100** to a network. The network is, for example, a local area network (LAN) and a wide area network (WAN). A method of connection to the network by the network interface **1060** may be wireless connection or wired connection. The image processing apparatus **100** may communicate with the camera **200** via the network interface **1060**. When the database **110** is located outside the image processing apparatus **100**, the image processing apparatus **100** may be connected to the database **110** via the network interface **1060**.

FIGS. 3 to 5 illustrate operations of the image processing apparatus **100** according to the present example embodiment. FIG. 3 illustrates a flow from image acquisition to search processing in the image processing apparatus **100**, FIG. 4 illustrates a flow of classification processing (S104) in FIG. 3, and FIG. 5 illustrates a flow of the search processing (S105) in FIG. 3.

As illustrated in FIG. 3, the image processing apparatus **100** acquires an image from the camera **200** (S101). The image acquisition unit **101** acquires an image in which a person is captured for performing classification and a search based on a skeleton structure, and stores the acquired image in the database **110**. For example, the image acquisition unit **101** acquires a plurality of images captured in a predetermined surveillance period, and performs the following processing on all persons included in the plurality of images.

Subsequently, the image processing apparatus **100** detects a skeleton structure of a person, based on the acquired image of the person (S102). FIG. 6 illustrates a detection example of skeleton structures. As illustrated in FIG. 6, a plurality of persons are included in an image acquired from a surveillance camera or the like, and a skeleton structure is detected for each of the persons included in the image.

FIG. 7 illustrates a skeleton structure of a human model **300** detected at this time, and FIGS. 8 to 10 each illustrate a detection example of the skeleton structure. The skeleton structure detection unit **102** detects the skeleton structure of the human model (two-dimensional skeleton model) **300** as in FIG. 7 from a two-dimensional image by using a skeleton estimation technique such as Open Pose. The human model **300** is a two-dimensional model formed of a keypoint such as a joint of a person and a bone connecting keypoints.

For example, the skeleton structure detection unit **102** extracts a feature point that may be a keypoint from an image, refers to information acquired by performing machine learning on the image of the keypoint, and detects each keypoint of a person. In the example illustrated in FIG. 7, as a keypoint of a person, a head A1, a neck A2, a right shoulder A31, a left shoulder A32, a right elbow A41, a left elbow A42, a right hand A51, a left hand A52, a right waist A61, a left waist A62, a right knee A71, a left knee A72, a right foot A81, and a left foot A82 are detected. Furthermore, as a bone of the person connecting the keypoints, detected are a bone B1 connecting the head A1 and the neck A2, a bone B21 connecting the neck A2 and the right shoulder A31, a bone B22 connecting the neck A2 and the left shoulder A32, a bone B31 connecting the right shoulder A31 and the right elbow A41, a bone B32 connecting the left shoulder A32 and the left elbow A42, a bone B41 connecting the right elbow A41 and the right hand A51, a bone B42 connecting the left elbow A42 and the left hand A52, a bone B51 connecting the neck A2 and the right waist A61, a bone B52 connecting the neck A2 and the left waist A62, a bone B61 connecting the right waist A61 and the right knee A71, a bone B62 connecting the left waist A62 and the left knee A72, a bone B71 connecting the right knee A71 and the right foot A81, and a bone B72 connecting the left knee A72 and the left foot A82. The skeleton structure detection unit **102** stores the detected skeleton structure of the person in the database **110**.

FIG. 8 is an example of detecting a person in an upright state. In FIG. 8, an image of the upright person is captured from the front, the bone B1, the bone B51 and the bone B52, the bone B61 and the bone B62, and the bone B71 and the bone B72 that are viewed from the front are each detected

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without overlapping, and the bone B61 and the bone B71 of a right leg are bent slightly more than the bone B62 and the bone B72 of a left leg.

FIG. 9 is an example of detecting a person in a squatting state. In FIG. 9, an image of the squatting person is captured from a right side, the bone B1, the bone B51 and the bone B52, the bone B61 and the bone B62, and the bone B71 and the bone B72 that are viewed from the right side are each detected, and the bone B61 and the bone B71 of a right leg and the bone B62 and the bone B72 of a left leg are greatly bent and also overlap.

FIG. 10 is an example of detecting a person in a sleeping state. In FIG. 10, an image of the sleeping person is captured diagonally from the front left, the bone B1, the bone B51 and the bone B52, the bone B61 and the bone B62, and the bone B71 and the bone B72 that are viewed diagonally from the front left are each detected, and the bone B61 and the bone B71 of a right leg, and the bone B62 and the bone B72 of a left leg are bent and also overlap.

Subsequently, as illustrated in FIG. 3, the image processing apparatus 100 computes a feature value of the detected skeleton structure (S103). For example, when a height and an area of a skeleton region are set a feature value, the feature value computation unit 103 extracts a region including the skeleton structure and acquires a height (pixel count) and an area (pixel area) of the region. The height and the area of the skeleton region are acquired from coordinates of an end portion of the extracted skeleton region and coordinates of a keypoint of the end portion. The feature value computation unit 103 stores the acquired feature value of the skeleton structure in the database 110. Note that, the feature value of the skeleton structure is also used as pose information indicating a pose of the person.

In the example in FIG. 8, a skeleton region including all of the bones is extracted from the skeleton structure of the upright person. In this case, an upper end of the skeleton region is the keypoint A1 of the head, a lower end of the skeleton region is the keypoint A82 of the left foot, a left end of the skeleton region is the keypoint A41 of the right elbow, and a right end of the skeleton region is the keypoint A52 of the left hand. Thus, a height of the skeleton region is acquired from a difference in Y coordinate between the keypoint A1 and the keypoint A82. Further, a width of the skeleton region is acquired from a difference in X coordinate between the keypoint A41 and the keypoint A52, and an area is acquired from the height and the width of the skeleton region.

In the example in FIG. 9, a skeleton region including all of the bones is extracted from the skeleton structure of the squatting person. In this case, an upper end of the skeleton region is the keypoint A1 of the head, a lower end of the skeleton region is the keypoint A81 of the right foot, a left end of the skeleton region is the keypoint A61 of the right waist, and a right end of the skeleton region is the keypoint A51 of the right hand. Thus, a height of the skeleton region is acquired from a difference in Y coordinate between the keypoint A1 and the keypoint A81. Further, a width of the skeleton region is acquired from a difference in X coordinate between the keypoint A61 and the keypoint A51, and an area is acquired from the height and the width of the skeleton region.

In the example in FIG. 10, a skeleton region including all of the bones is extracted from the skeleton structure of the sleeping person lying along the left-right direction of the image. In this case, an upper end of the skeleton region is the keypoint A32 of the left shoulder, a lower end of the skeleton region is the keypoint A52 of the left hand, a left end of the

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skeleton region is the keypoint A51 of the right hand, and a right end of the skeleton region is the keypoint A82 of the left foot. Thus, a height of the skeleton region is acquired from a difference in Y coordinate between the keypoint A32 and the keypoint A52. Further, a width of the skeleton region is acquired from a difference in X coordinate between the keypoint A51 and the keypoint A82, and an area is acquired from the height and the width of the skeleton region.

Subsequently, as illustrated in FIG. 3, the image processing apparatus 100 performs classification processing (S104). In the classification processing, as illustrated in FIG. 4, the classification unit 104 computes a degree of similarity of the computed feature value of the skeleton structure (S111), and classifies the skeleton structure based on the degree of similarity (S112). The classification unit 104 acquires a degree of similarity among all of the skeleton structures that are subjects to be classified and are stored in the database 110, and classifies skeleton structures (poses) having a highest degree of similarity in the same cluster (performs clustering). Furthermore, classification is performed by acquiring a degree of similarity between classified clusters, and classification is repeated until the number of clusters becomes a predetermined number. FIG. 11 illustrates an image of a classification result of feature values of skeleton structures. FIG. 11 is an image of a cluster analysis by two-dimensional classification elements, and two classification elements are, for example, a height of a skeleton region and an area of the skeleton region, or the like. In FIG. 11, as a result of classification, feature values of a plurality of skeleton structures are classified into three clusters C1 to C3. The clusters C1 to C3 are associated with poses such as a standing pose, a sitting pose, and a sleeping pose, respectively, for example, and skeleton structures (persons) are classified for each similar pose.

In the present example embodiment, various classification methods can be used by performing classification, based on a feature value of a skeleton structure of a person. Note that, a classification method may be preset, or any classification method may be able to be set by a user. Further, classification may be performed by the same method as a search method described below. In other words, classification may be performed by a classification condition similar to a search condition. For example, the classification unit 104 performs classification by the following classification methods. Any classification method may be used, or any selected classification methods may be combined.

(Classification Method 1) Classification by a Plurality of Hierarchical Levels

Classification is performed by combining, in a hierarchical manner, classification by a skeleton structure of a whole body, classification by a skeleton structure of an upper body and a lower body, classification by a skeleton structure of an arm and a leg, and the like. In other words, classification may be performed based on a feature value of a first portion and a second portion of a skeleton structure, and, furthermore, classification may be performed by assigning weights to the feature value of the first portion and the second portion.

(Classification Method 2) Classification by a Plurality of Images Along Time Series

Classification is performed based on a feature value of a skeleton structure in a plurality of images successive in time series. For example, classification may be performed based on a cumulative value by accumulating a feature value in a time series direction. Furthermore, classification may be

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performed based on a change (change value) in a feature value of a skeleton structure in a plurality of successive images.

(Classification Method 3) Classification by Ignoring the Left and the Right of a Skeleton Structure

Classification is performed on an assumption that skeleton structures in which a right side and a left side are reversed are the same skeleton structure.

Furthermore, the classification unit **104** displays a classification result of the skeleton structure (**S113**). The classification unit **104** acquires a necessary image of a skeleton structure and a person from the database **110**, and displays, on the display unit **107**, the skeleton structure and the person for each similar pose (cluster) as a classification result. FIG. **12** illustrates a display example when poses are classified into three. For example, as illustrated in FIG. **12**, pose regions **WA1** to **WA3** for each pose are displayed on a display window **W1**, and a skeleton structure and a person (image) of each associated pose are displayed in the pose regions **WA1** to **WA3**. The pose region **WA1** is, for example, a display region of a standing pose, and displays a skeleton structure and a person that are classified into the cluster **C1** and are similar to the standing pose. The pose region **WA2** is, for example, a display region of a sitting pose, and displays a skeleton structure and a person that are classified into the cluster **C2** and are similar to the sitting pose. The pose region **WA3** is, for example, a display region of a sleeping pose, and displays a skeleton structure and a person that are classified into the cluster **C3** and are similar to the sleeping pose.

Subsequently, as illustrated in FIG. **3**, the image processing apparatus **100** performs the search processing (**S105**). In the search processing, as illustrated in FIG. **5**, the search unit **105** receives an input of a search condition (**S121**), and searches for a skeleton structure, based on the search condition (**S122**). The search unit **105** receives, from the input unit **106**, an input of a search query being the search condition in response to an operation of a user. When the search query is input from a classification result, for example, in the display example in FIG. **12**, a user specifies (selects), from among the pose regions **WA1** to **WA3** displayed on the display window **W1**, a skeleton structure of a pose desired to be searched for. Then, with the skeleton structure specified by the user as the search query, the search unit **105** searches for a skeleton structure having a high degree of similarity of a feature value from among all of the skeleton structures that are subjects to be searched and are stored in the database **110**. The search unit **105** computes a degree of similarity between a feature value of the skeleton structure being the search query and a feature value of the skeleton structure being the subject to be searched, and extracts a skeleton structure having the computed degree of similarity higher than a predetermined threshold value. The feature value of the skeleton structure being the search query may use a feature value being computed in advance, or may use a feature value being acquired during a search. Note that, the search query may be input by moving each portion of a skeleton structure in response to an operation of the user, or a pose demonstrated by the user in front of a camera may be set as the search query.

In the present example embodiment, similarly to the classification methods, various search methods can be used by performing a search, based on a feature value of a skeleton structure of a person. Note that, a search method may be preset, or any search method may be able to be set by a user. For example, the search unit **105** performs a search by the following search methods. Any search method may be

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used, or any selected search methods may be combined. A search may be performed by combining a plurality of search methods (search conditions) by a logical expression (for example, AND (conjunction), OR (disjunction), NOT (negation)). For example, a search may be performed by setting “(pose with a right hand up) AND (pose with a left foot up)” as a search condition.

(Search Method 1) A search only by a feature value in the height direction By performing a search by using only a feature value in the height direction of a search person, an influence of a change in the horizontal direction of a person can be suppressed, and robustness improves with respect to a change in orientation of the person and body shape of the person. For example, as in skeleton structures **501** to **503** in FIG. **13**, even when there is difference in an orientation or a body shape of a person, a feature value in the height direction does not greatly change. Thus, in the skeleton structures **501** to **503**, it can be decided, at a time of a search (at a time of classification), that poses are the same.

(Search Method 2) When a part of a body of a person is hidden in a partial search image, a search is performed by using only information about a recognizable portion. For example, as in skeleton structures **511** and **512** in FIG. **14**, even when a keypoint of a left foot cannot be detected due to the left foot being hidden, a search can be performed by using a feature value of another detected keypoint. Thus, in the skeleton structures **511** and **512**, it can be decided, at a time of a search (at a time of classification), that poses are the same. In other words, classification and a search can be performed by using a feature value of some of keypoints instead of all keypoints. In an example of skeleton structures **521** and **522** in FIG. **15**, although orientations of both feet are different, it can be decided that poses are the same by setting a feature value of keypoints (**A1**, **A2**, **A31**, **A32**, **A41**, **A42**, **A51**, and **A52**) of an upper body as a search query. Further, a search may be performed by assigning a weight to a portion (feature point) desired to be searched, or a threshold value of a similarity degree determination may be changed. When a part of a body is hidden, a search may be performed by ignoring the hidden portion, or a search may be performed by taking the hidden portion into consideration. By performing a search also including a hidden portion, a pose in which the same portion is hidden can be searched.

(Search Method 3) Search by Ignoring the Left and the Right of a Skeleton Structure

A search is performed on an assumption that skeleton structures in which a right side and a left side are reversed are the same skeleton structure. For example, as in skeleton structures **531** and **532** in FIG. **16**, a pose with a right hand up and a pose with a left hand up can be searched (classified) as the same pose. In the example in FIG. **16**, in the skeleton structure **531** and the skeleton structure **532**, although positions of the keypoint **A51** of the right hand, the keypoint **A41** of the right elbow, the keypoint **A52** of the left hand, and the keypoint **A42** of the left elbow are different, positions of the other keypoints are the same. When the keypoints of one of the skeleton structures, of the keypoint **A51** of the right hand and the keypoint **A41** of the right elbow of the skeleton structure **531** and the keypoint **A52** of the left hand and the keypoint **A42** of the left elbow of the skeleton structure **532**, are reversed, the keypoints have the same positions of the keypoints of the other skeleton structure. When the keypoints of one of the skeleton structures, of the keypoint **A52** of the left hand and the keypoint **A42** of the left elbow of the skeleton structure **531** and the keypoint **A51** of the right hand and the keypoint **A41** of the right elbow of the skeleton

structure **532**, are reversed, the keypoints have the same positions of the keypoints of the other skeleton structure. Thus, it is decided that poses are the same.

(Search Method 4) A search by a Feature Value in the Vertical Direction and the Horizontal Direction

After a search is performed only with a feature value of a person in the vertical direction (Y-axis direction), the acquired result is further searched by using a feature value of the person in the horizontal direction (X-axis direction). (Search Method 5) A Search by a Plurality of Images Along Time Series

A search is performed based on a feature value of a skeleton structure in a plurality of images successive in time series. For example, a search may be performed based on a cumulative value by accumulating a feature value in a time series direction. Furthermore, a search may be performed based on a change (change value) in a feature value of a skeleton structure in a plurality of successive images.

Furthermore, the search unit **105** displays a search result of the skeleton structure (**S123**). The search unit **105** acquires a necessary image of a skeleton structure and a person from the database **110**, and displays, on the display unit **107**, the skeleton structure and the person acquired as a search result. For example, when a plurality of search queries (search conditions) are specified, a search result is displayed for each of the search queries. FIG. **17** illustrates a display example when a search is performed by three search queries (poses). For example, as illustrated in FIG. **17**, in a display window **W2**, skeleton structures and persons of search queries **Q10**, **Q20**, and **Q30** specified are displayed at a left end portion, and skeleton structures and persons of search results **Q11**, **Q21**, and **Q31** of the search queries are displayed side by side on the right side of the search queries **Q10**, **Q20**, and **Q30**.

An order in which search results are displayed side by side from a search query may be an order in which a corresponding skeleton structure is found, or may be decreasing order of a degree of similarity. When a search is performed by assigning a weight to a portion (feature point) in a partial search, display may be performed in an order of a degree of similarity computed by assigning a weight. Display may be performed in an order of a degree of similarity computed only from a portion (feature point) selected by a user. Further, display may be performed by cutting, for a certain period of time, images (frames) in time series before and after an image (frame) that is a search result.

(Search Method 6) In the present search method, the database **110** stores a video. The processing described above is performed on each of a plurality of frame images constituting a video. Then, the image processing apparatus **100** searches for a video similar to a video (hereinafter described as a query video) to be a query from the database **110**.

FIG. **40** is a diagram illustrating a functional configuration of the search unit **105** according to the present search method. The search unit **105** illustrated in FIG. **40** includes a query acquisition unit **610**, a query frame selection unit **620**, and a video selection unit **630**.

The query acquisition unit **610** acquires the query video described above. In the following description, a frame image included in the query video is described as a first frame image. In other words, a query image includes a plurality of first frame images.

The query frame selection unit **620** selects a plurality of query frames from among the plurality of first frame images. The query frame is used when a video similar to the query video is selected from a plurality of videos being subjects to be searched. The number of the query frames may be two or

more, but a greater number is more preferable. A selection reference of the query frame will be described below.

Note that, another frame image may be present between a certain query frame image and a next query frame image. In any selection reference described below, the number of frame images (or time) between query frames preferably falls within a reference. The reference used herein is, for example, equal to or more than one frame and equal to or less than 10 frames, or equal to or more than 0.025 second and equal to or less than one second.

The video selection unit **630** selects a video by using the plurality of query frames. At this time, the video selection unit **630** uses References (1) and (2) below as a condition for selecting the video.

Reference (1) A condition that a “similar frame image whose degree of similarity to a query frame satisfies a first reference is present” is satisfied for at least two query frames.

Reference (2) When a query frame associated with each of at least two similar frame images is arranged in the same order as that of the at least two similar frame images, the arrangement order coincides with an arrangement order of at least two query frames in a query video.

First, (1) described above will be described. When a first video and a second video are similar to each other, each of at least two frame images included in the first video is assumed to be similar to any frame image included in the second video. (1) described above corresponds to this.

Next, (2) described above will be described. It needs to be decided that the first video and the second video are different from each other even in a case where a plurality of frame images are similar therebetween, when appearance orders of the plurality of frame images do not coincide with each other. Reference (2) requires appearance orders of a plurality of frame images to coincide with each other.

References (1) and (2) described above can be replaced with the following description.

Reference (1)' A first similar frame image whose degree of similarity to at least a first query frame satisfies a reference and a second similar frame image whose degree of similarity to a second query frame satisfies the reference are included.

Reference (2)' An order of the first similar frame image and the second similar frame image is the same as an order of the first query frame and the second query frame.

Note that, another frame image may be present between the first similar frame image and the second similar frame image. In this case, a fact that the number of frame images (or time) between a first similar frame and a second similar frame falls within a reference may be further added to Reference (2) (or Reference (2)'). The reference used herein is, for example, equal to or more than one frame and equal to or less than 10 frames, or equal to or more than 0.025 second and equal to or less than one second.

A video to which the present search method is applicable is not limited to a video including a pose of a person. However, when a video including a pose of a person is searched, a degree of similarity of a pose of a person can be used as a degree of similarity between frame images.

When a degree of similarity of a pose of a person is used as a degree of similarity between frame images, a condition for selection as a query frame preferably includes a condition that an “information amount related to a person included in a query frame satisfies a second reference”. In other words, information related to a person can be acquired by an image analysis, but a query frame needs a great amount of the information amount to some extent. One example of the second reference is that the number of

keypoints (joints) included in a query frame satisfies a reference. The reference is set as a value of 30% or more (for example, five out of 14) of a total number of keypoints, for example. Herein, at least one or more keypoints associated with a hand and a foot are preferably included. Note that, a portion associated with a neck, both shoulders, and a head among keypoints is less likely to be lost. When the number of keypoints is small, an information amount related to a pose of a person decreases, and thus search accuracy of a video decreases.

FIG. 41 is a diagram for describing one example of a method for selecting a similar frame by the video selection unit 630. In the example illustrated in FIG. 41, a pose of a person is defined by a feature value (parameter) of a skeleton structure described above. Then, the video selection unit 630 selects a similar frame by using a distance in a space (hereinafter described as a feature value space) defined by the feature value. Specifically, the video selection unit 630 selects, as a similar frame image, a frame image having a distance from a query frame within a first reference in the feature value space. Note that, the video selection unit 630 sets the first reference according to an input from a user. However, the video selection unit 630 may use a value predetermined as the first reference.

FIG. 42 is a flowchart illustrating one example of processing performed by the search unit 105 according to the present search method. FIG. 43 is a diagram for describing the processing illustrated in FIG. 42.

First, the query acquisition unit 610 acquires a query video (step S300). As one example, the query acquisition unit 610 acquires, as a query video, a video specified by a user. The user may select a query video from among videos stored in the database 110, or may cause the search unit 105 to acquire a query video from an external apparatus or a storage medium.

Next, the query frame selection unit 620 selects a plurality of query frames from a plurality of first frame images included in the query video. In the example illustrated in FIG. 42, the query frame selection unit 620 selects a plurality of query frames according to an input of the user (step S302).

Specifically, as illustrated in FIG. 43, the query frame selection unit 620 arranges the plurality of first frame images (preferably all the first frame images) included in the query video in an appearance order, and displays the plurality of first frame images on the display unit 107. Then, the user selects the first frame image needed to be a query frame by using an input device such as a mouse.

Next, the video selection unit 630 selects a video similar to the query video by using the plurality of query frames selected in step S302 (step S304). The selection reference is as described by using FIGS. 40 and 41. Note that, also in the diagram illustrated in FIG. 43, the selected video satisfies References (1) and (2) described above, and also satisfies Reference (1)' and (2)'.

Note that, when the query video is not stored in the database 110, the skeleton structure detection unit 102 and the feature value computation unit 103 process the query frames and compute a feature value of a skeleton structure. Then, the video selection unit 630 selects a video similar to the query video by using the feature value of the skeleton structure.

Subsequently, the video selection unit 630 stores a selection result in the database 110. Herein, the video selection unit 630 may store the selected video itself, or may associ-

ate, with the video already stored in the database 110, a flag indicating that the image is similar to the query video (step S306).

FIG. 44 is a flowchart illustrating a modification example of FIG. 42. The present modification example is similar to the example illustrated in FIG. 42 except for a point that the query frame selection unit 620 selects a plurality of query images according to a predetermined rule (hereinafter described as a selection rule) (step S303).

FIG. 45 is a diagram for describing a first example of the selection rule of a query frame. In the example illustrated in FIG. 45, the query frame selection unit 620 selects a plurality of query frames from a plurality of first frame images at predetermined intervals. Herein, the query frame selection unit 620 selects an n-th first frame image as a first query frame. Herein, n may be a preset integer, or may be an integer set by a user input. Further, the predetermined interval is, for example, equal to or more than two frames and equal to or less than 10 frames, or equal to or more than 0.05 second and equal to or less than one second.

FIG. 46 is a diagram for describing a second example of the selection rule of a query frame. In the example illustrated in FIG. 46, after a first query frame is determined, the query frame selection unit 620 selects, as a query frame following the first query frame, a first frame image having a change amount from the first query frame equal to or more than a third reference. Then, the query frame selection unit 620 repeats the processing. In this way, poses of a person included in adjacent query frames are different from each other to some extent. Therefore, search accuracy of a video by the search unit 105 can be increased while suppressing an increase in a query frame.

Herein, the third reference may be set according to an input from a user, or may be set by using a plurality of the first query frames.

FIG. 47 is a diagram for describing a method for setting the third reference by using the plurality of first query frames by the query frame selection unit 620. In the example illustrated in FIG. 47, the third reference is a distance in the feature value space. Then, there are two specific examples below.

In the first example, the query frame selection unit 620 determines a maximum distance between two query frames in the feature value space, and sets the third reference by using the maximum distance. As one example, the query frame selection unit 620 sets the third reference to a value less than the maximum distance by using a function with the maximum distance as a variable. As one example, the query frame selection unit 620 sets the third reference by multiplying the maximum distance by a coefficient less than one or subtracting a predetermined value from the maximum distance.

In the second example, the query frame selection unit 620 sets the third reference by using a result acquired by performing statistical processing on a distance between two first frames adjacent in terms of time. For example, the query frame selection unit 620 may set a medium value, an average value, or a mode of the distances as the third reference, or may set the third reference by using a function with at least one of a medium value, an average value, and a mode as a variable.

Note that, also in the examples illustrated in FIGS. 46 and 47, the query frame selection unit 620 selects an n-th first frame image as a first query frame similarly to the example illustrated in FIG. 45.

As described above, in the present example embodiment, a skeleton structure of a person can be detected from a

two-dimensional image, and classification and a search can be performed based on a feature value of the detected skeleton structure. In this way, classification can be performed for each similar pose having a high degree of similarity, and a similar pose having a high degree of similarity to a search query (search key) can be searched. By classifying similar poses from an image and displaying the similar poses, a user can recognize a pose of a person in the image without specifying a pose and the like. Since the user can specify a pose being a search query from a classification result, a desired pose can be searched for even when a pose desired to be searched for by a user is not recognized in detail in advance. For example, since classification and a search can be performed with a whole or a part of a skeleton structure of a person and the like as a condition, flexible classification and a flexible search can be performed.

Further, according to the search method 6, a video similar to a query video can be accurately searched for. Further, even when frame rates are different or production times are different between a query video and a video to be a subject to be searched, a search for a video can be performed.

Example Embodiment 2

An example embodiment 2 will be described below with reference to the drawings. In the present example embodiment, a specific example of the feature value computation in the example embodiment 1 will be described. In the present example embodiment, a feature value is acquired by normalization by using a height of a person. The other points are similar to those in the example embodiment 1.

FIG. 18 illustrates a configuration of an image processing apparatus 100 according to the present example embodiment. As illustrated in FIG. 18, the image processing apparatus 100 further includes a height computation unit 108 in addition to the configuration in the example embodiment 1. Note that, a feature value computation unit 103 and the height computation unit 108 may serve as one processing unit.

The height computation unit (height estimation unit) 108 computes (estimates) an upright height (referred to as a height pixel count) of a person in a two-dimensional image, based on a two-dimensional skeleton structure detected by a skeleton structure detection unit 102. It can be said that the height pixel count is a height of a person in a two-dimensional image (a length of a whole body of a person on a two-dimensional image space). The height computation unit 108 acquires a height pixel count (pixel count) from a length (length on the two-dimensional image space) of each bone of a detected skeleton structure.

In the following examples, specific examples 1 to 3 are used as a method for acquiring a height pixel count. Note that, any method of the specific examples 1 to 3 may be used, or a plurality of any selected methods may be combined and used. In the specific example 1, a height pixel count is acquired by adding up lengths of bones from a head to a foot among bones of a skeleton structure. When the skeleton structure detection unit 102 (skeleton estimation technique) does not output a top of a head and a foot, a correction can be performed by multiplication by a constant as necessary. In the specific example 2, a height pixel count is computed by using a human model indicating a relationship between a length of each bone and a length of a whole body (a height on the two-dimensional image space). In the specific example 3, a height pixel count is computed by fitting (applying) a three-dimensional human model to a two-dimensional skeleton structure.

The feature value computation unit 103 according to the present example embodiment is a normalization unit that normalizes a skeleton structure (skeleton information) of a person, based on a computed height pixel count of the person. The feature value computation unit 103 stores a feature value (normalization value) of the normalized skeleton structure in a database 110. The feature value computation unit 103 normalizes, by the height pixel count, a height on an image of each keypoint (feature point) included in the skeleton structure. In the present example embodiment, for example, a height direction is an up-down direction (Y-axis direction) in a two-dimensional coordinate (X-Y coordinate) space of an image. In this case, a height of a keypoint can be acquired from a value (pixel count) of a Y coordinate of the keypoint. Alternatively, a height direction may be a direction (vertical projection direction) of a vertical projection axis in which a direction of a vertical axis perpendicular to the ground (reference surface) in a three-dimensional coordinate space in a real world is projected in the two-dimensional coordinate space. In this case, a height of a keypoint can be acquired from a value (pixel count) along a vertical projection axis, the vertical projection axis being acquired by projecting an axis perpendicular to the ground in the real world to the two-dimensional coordinate space, based on a camera parameter. Note that, the camera parameter is a capturing parameter of an image, and, for example, the camera parameter is a pose, a position, a capturing angle, a focal distance, and the like of a camera 200. The camera 200 captures an image of an object whose length and position are clear in advance, and a camera parameter can be acquired from the image. A strain may occur at both ends of the captured image, and the vertical direction in the real world and the up-down direction in the image may not match. In contrast, an extent that the vertical direction in the real world is tilted in an image is clear by using a parameter of a camera that captures the image. Thus, a feature value of a keypoint can be acquired in consideration of a difference between the real world and the image by normalizing, by a height, a value of the keypoint along a vertical projection axis projected in the image, based on the camera parameter. Note that, a left-right direction (a horizontal direction) is a direction (X-axis direction) of left and right in a two-dimensional coordinate (X-Y coordinate) space of an image, or is a direction in which a direction parallel to the ground in the three-dimensional coordinate space in the real world is projected to the two-dimensional coordinate space.

FIGS. 19 to 23 illustrate operations of the image processing apparatus 100 according to the present example embodiment. FIG. 19 illustrates a flow from image acquisition to search processing in the image processing apparatus 100, FIGS. 20 to 22 illustrate flows of specific examples 1 to 3 of height pixel count computation processing (S201) in FIG. 19, and FIG. 23 illustrates a flow of normalization processing (S202) in FIG. 19.

As illustrated in FIG. 19, in the present example embodiment, the height pixel count computation processing (S201) and the normalization processing (S202) are performed as the feature value computation processing (S103) in the example embodiment 1. The other points are similar to those in the example embodiment 1.

The image processing apparatus 100 performs the height pixel count computation processing (S201), based on a detected skeleton structure, after the image acquisition (S101) and skeleton structure detection (S102). In this example, as illustrated in FIG. 24, a height of a skeleton structure of an upright person in an image is a height pixel

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count (h), and a height of each keypoint of the skeleton structure in the state of the person in the image is a keypoint height (yi). Hereinafter, the specific examples 1 to 3 of the height pixel count computation processing will be described.

Specific Example 1

In the specific example 1, a height pixel count is acquired by using a length of a bone from a head to a foot. In the specific example 1, as illustrated in FIG. 20, the height computation unit 108 acquires a length of each bone (S211), and adds up the acquired length of each bone (S212).

The height computation unit 108 acquires a length of a bone from a head to a foot of a person on a two-dimensional image, and acquires a height pixel count. In other words, each length (pixel count) of a bone B1 (length L1), a bone B51 (length L21), a bone B61 (length L31), and a bone B71 (length L41), or the bone B1 (length L1), a bone B52 (length L22), a bone B62 (length L32), and a bone B72 (length L42) among bones in FIG. 24 is acquired from the image in which the skeleton structure is detected. A length of each bone can be acquired from coordinates of each keypoint in the two-dimensional image. A value acquired by multiplying, by a correction constant, $L+L21+L31+L41$ or $L1+L22+L32+L42$, acquired by adding them up, is computed as the height pixel count (h). When both values can be computed, a longer value is set as the height pixel count, for example. In other words, each bone has a longest length in an image when being captured from the front, and is displayed to be short when being tilted in a depth direction with respect to a camera. Therefore, it is conceivable that a longer bone has a higher possibility of being captured from the front, and has a value closer to a true value. Thus, a longer value is preferably selected.

In an example in FIG. 25, the bone B1, the bone B51 and the bone B52, the bone B61 and the bone B62, and the bone B71 and the bone B72 are each detected without overlapping. $L1+L21+L31+L41$ and $L1+L22+L32+L42$ that are a total of the bones are acquired, and, for example, a value acquired by multiplying, by a correction constant, $L1+L22+L32+L42$ on a left leg side having a greater length of the detected bones is set as the height pixel count.

In an example in FIG. 26, the bone B1, the bone B51 and the bone B52, the bone B61 and the bone B62, and the bone B71 and the bone B72 are each detected, and the bone B61 and the bone B71 of a right leg, and the bone B62 and the bone B72 of a left leg overlap. $L1+L21+L31+L41$ and $L1+L22+L32+L42$ that are a total of the bones are acquired, and, for example, a value acquired by multiplying, by a correction constant, $L1+L21+L31+L41$ on a right leg side having a greater length of the detected bones is set as the height pixel count.

In an example in FIG. 27, the bone B1, the bone B51 and the bone B52, the bone B61 and the bone B62, and the bone B71 and the bone B72 are each detected, and the bone B61 and the bone B71 of the right leg and the bone B62 and the bone B72 of the left leg overlap. $L1+L21+L31+L41$ and $L1+L22+L32+L42$ that are a total of the bones are acquired, and, for example, a value acquired by multiplying, by a correction constant, $L1+L22+L32+L42$ on the left leg side having a greater length of the detected bones is set as the height pixel count.

In the specific example 1, since a height can be acquired by adding up lengths of bones from a head to a foot, a height pixel count can be acquired by a simple method. Further, since at least a skeleton from a head to a foot may be able to be detected by a skeleton estimation technique using

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machine learning, a height pixel count can be accurately estimated even when the entire person is not necessarily captured in an image as in a squatting state and the like.

Specific Example 2

In the specific example 2, a height pixel count is acquired by using a two-dimensional skeleton model indicating a relationship between a length of a bone included in a two-dimensional skeleton structure and a length of a whole body of a person on a two-dimensional image space.

FIG. 28 is a human model (two-dimensional skeleton model) 301 that is used in the specific example 2 and indicates a relationship between a length of each bone on the two-dimensional image space and a length of a whole body on the two-dimensional image space. As illustrated in FIG. 28, a relationship between a length of each bone of an average person and a length of a whole body (a proportion of a length of each bone to a length of a whole body) is associated with each bone of the human model 301. For example, a length of the bone B1 of a head is the length of the whole body $\times$ 0.2 (20%), a length of the bone B41 of a right hand is the length of the whole body $\times$ 0.15 (15%), and a length of the bone B71 of the right leg is the length of the whole body $\times$ 0.25 (25%). Information about such a human model 301 is stored in the database 110, and thus an average length of a whole body can be acquired from a length of each bone. In addition to a human model of an average person, a human model may be prepared for each attribute of a person such as age, sex, and nationality. In this way, a length (height) of a whole body can be appropriately acquired according to an attribute of a person.

In the specific example 2, as illustrated in FIG. 21, the height computation unit 108 acquires a length of each bone (S221). The height computation unit 108 acquires a length of all bones (length on the two-dimensional image space) in a detected skeleton structure. FIG. 29 is an example of capturing an image of a person in a squatting state diagonally from rear right and detecting a skeleton structure. In this example, since a face and a left side surface of a person are not captured, a bone of a head and bones of a left arm and a left hand cannot be detected. Thus, each length of bones B21, B22, B31, B41, B51, B52, B61, B62, B71, and B72 that are detected is acquired.

Subsequently, as illustrated in FIG. 21, the height computation unit 108 computes a height pixel count from a length of each bone, based on a human model (S222). The height computation unit 108 refers to the human model 301 indicating a relationship between lengths of each bone and a whole body as in FIG. 28, and acquires a height pixel count from the length of each bone. For example, since a length of the bone B41 of the right hand is the length of the whole body $\times$ 0.15, a height pixel count based on the bone B41 is acquired from the length of the bone B41/0.15. Further, since a length of the bone B71 of the right leg is the length of the whole body $\times$ 0.25, a height pixel count based on the bone B71 is acquired from the length of the bone B71/0.25.

The human model referred at this time is, for example, a human model of an average person, but a human model may be selected according to an attribute of a person such as age, sex, and nationality. For example, when a face of a person is captured in a captured image, an attribute of the person is identified based on the face, and a human model associated with the identified attribute is referred. An attribute of a person can be recognized from a feature of a face in an image by referring to information acquired by performing machine learning on a face for each attribute. Further, when

an attribute of a person cannot be identified from an image, a human model of an average person may be used.

Further, a height pixel count computed from a length of a bone may be corrected by a camera parameter. For example, when a camera is placed in a high position and performs capturing in such a way that a person is looked down, a horizontal length such as a bone of a shoulder width is not affected by a dip of the camera in a two-dimensional skeleton structure, but a vertical length such as a bone from a neck to a waist is reduced as a dip of the camera increases. Then, a height pixel count computed from the horizontal length such as a bone of a shoulder width tends to be greater than an actual height pixel count. Thus, when a camera parameter is used, an angle at which a person is looked down by the camera is clear, and thus a correction can be performed in such a way as to acquire a two-dimensional skeleton structure captured from the front by using information about the dip. In this way, a height pixel count can be more accurately computed.

Subsequently, as illustrated in FIG. 21, the height computation unit 108 computes an optimum value of the height pixel count (S223). The height computation unit 108 computes an optimum value of the height pixel count from the height pixel count acquired for each bone. For example, a histogram of a height pixel count acquired for each bone as illustrated in FIG. 30 is generated, and a great height pixel count is selected from among the height pixel counts. In other words, a longer height pixel count is selected from among a plurality of height pixel counts acquired based on a plurality of bones. For example, values in top 30% are regarded valid, and height pixel counts by the bones B71, B61, and B51 are selected in FIG. 30. An average of the selected height pixel counts may be acquired as an optimum value, or a greatest height pixel count may be set as an optimum value. Since a height is acquired from a length of a bone in a two-dimensional image, when the bone cannot be captured from the front, i.e., when the bone tilted in the depth direction as viewed from the camera is captured, a length of the bone is shorter than that captured from the front. Then, a value having a greater height pixel count has a higher possibility of being captured from the front than a value having a smaller height pixel count and is a more plausible value, and thus a greater value is set as an optimum value.

In the specific example 2, since a height pixel count is acquired based on a bone of a detected skeleton structure by using a human model indicating a relationship between lengths of a bone and a whole body on the two-dimensional image space, a height pixel count can be acquired from some of bones even when not all skeletons from a head to a foot can be acquired. Particularly, a height pixel count can be accurately estimated by adopting a greater value from among values acquired from a plurality of bones.

Specific Example 3

In the specific example 3, a skeleton vector of a whole body is acquired by fitting a two-dimensional skeleton structure to a three-dimensional human model (three-dimensional skeleton model) and using a height pixel count of the fit three-dimensional human model.

In the specific example 3, as illustrated in FIG. 22, the height computation unit 108 first computes a camera parameter, based on an image captured by the camera 200 (S231). The height computation unit 108 extracts an object whose length is clear in advance from a plurality of images captured by the camera 200, and acquires a camera parameter

from a size (pixel count) of the extracted object. Note that, a camera parameter may be acquired in advance, and the acquired camera parameter may be obtained as necessary.

Subsequently, the height computation unit 108 adjusts an arrangement and a height of a three-dimensional human model (S232). The height computation unit 108 prepares, for a detected two-dimensional skeleton structure, the three-dimensional human model for computing a height pixel count, and arranges the three-dimensional human model in the same two-dimensional image, based on the camera parameter. Specifically, a “relative positional relationship between a camera and a person in a real world” is determined from the camera parameter and the two-dimensional skeleton structure. For example, on the basis that a position of the camera has coordinates (0, 0, 0), coordinates (x, y, z) of a position in which a person stands (or sits) are determined. Then, by assuming an image captured when the three-dimensional human model is arranged in the same position (x, y, z) as that of the determined person, the two-dimensional skeleton structure and the three-dimensional human model are superimposed.

FIG. 31 is an example of capturing an image of a squatting person diagonally from front left and detecting a two-dimensional skeleton structure 401. The two-dimensional skeleton structure 401 includes two-dimensional coordinate information. Note that, all bones are preferably detected, but some of bones may not be detected. A three-dimensional human model 402 as in FIG. 32 is prepared for the two-dimensional skeleton structure 401. The three-dimensional human model (three-dimensional skeleton model) 402 is a model of a skeleton including three-dimensional coordinate information and having the same shape as that of the two-dimensional skeleton structure 401. Then, as in FIG. 33, the prepared three-dimensional human model 402 is arranged and superimposed on the detected two-dimensional skeleton structure 401. Further, the three-dimensional human model 402 is superimposed on the two-dimensional skeleton structure 401, and a height of the three-dimensional human model 402 is also adjusted to the two-dimensional skeleton structure 401.

Note that, the three-dimensional human model 402 prepared at this time may be a model in a state close to a pose of the two-dimensional skeleton structure 401 as in FIG. 33, or may be a model in an upright state. For example, the three-dimensional human model 402 with an estimated pose may be generated by using a technique for estimating a pose in a three-dimensional space from a two-dimensional image by using machine learning. A three-dimensional pose can be estimated from a two-dimensional image by learning information about a joint in the two-dimensional image and information about a joint in a three-dimensional space.

Subsequently, as illustrated in FIG. 22, the height computation unit 108 fits the three-dimensional human model to a two-dimensional skeleton structure (S233). As in FIG. 34, the height computation unit 108 deforms the three-dimensional human model 402 in such a way that poses of the three-dimensional human model 402 and the two-dimensional skeleton structure 401 match in a state where the three-dimensional human model 402 is superimposed on the two-dimensional skeleton structure 401. In other words, a height, an orientation of a body, and an angle of a joint of the three-dimensional human model 402 are adjusted, and optimization is performed in such a way as to eliminate a difference from the two-dimensional skeleton structure 401. For example, by rotating a joint of the three-dimensional human model 402 in a movable range of a person and also rotating the entire three-dimensional human model 402, the

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entire size is adjusted. Note that, fitting (application) between a three-dimensional human model and a two-dimensional skeleton structure is performed on a two-dimensional space (two-dimensional coordinates). In other words, a three-dimensional human model is mapped in the two-dimensional space, and the three-dimensional human model is optimized for a two-dimensional skeleton structure in consideration of a change of the deformed three-dimensional human model in the two-dimensional space (image).

Subsequently, as illustrated in FIG. 22, the height computation unit 108 computes a height pixel count of the fit three-dimensional human model (S234). As in FIG. 35, when there is no difference between the three-dimensional human model 402 and the two-dimensional skeleton structure 401 and poses match, the height computation unit 108 acquires a height pixel count of the three-dimensional human model 402 in that state. With the optimized three-dimensional human model 402 in an upright state, a length of a whole body on the two-dimensional space is acquired based on a camera parameter. For example, a height pixel count is computed from lengths (pixel counts) of bones from a head to a foot when the three-dimensional human model 402 is upright. Similarly to the specific example 1, the lengths of the bones from the head to the foot of the three-dimensional human model 402 may be added up.

In the specific example 3, a height pixel count is acquired based on a three-dimensional human model by fitting the three-dimensional human model to a two-dimensional skeleton structure, based on a camera parameter, and thus the height pixel count can be accurately estimated even when all bones are not captured at the front, i.e., when an error is great due to all bones being captured on a slant.

<Normalization Processing> As illustrated in FIG. 19, the image processing apparatus 100 performs the normalization processing (S202) after the height pixel count computation processing. In the normalization processing, as illustrated in FIG. 23, the feature value computation unit 103 computes a keypoint height (S241). The feature value computation unit 103 computes a keypoint height (pixel count) of all keypoints included in the detected skeleton structure. The keypoint height is a length (pixel count) in the height direction from a lowest end (for example, a keypoint of any foot) of the skeleton structure to the keypoint. Herein, as one example, the keypoint height is acquired from a Y coordinate of the keypoint in an image. Note that, as described above, the keypoint height may be acquired from a length along a vertical projection axis based on a camera parameter. For example, in the example in FIG. 24, a height (yi) of a keypoint A2 of a neck is a value acquired by subtracting a Y coordinate of a keypoint A81 of a right foot or a keypoint A82 of a left foot from a Y coordinate of the keypoint A2.

Subsequently, the feature value computation unit 103 determines a reference point for normalization (S242). The reference point is a point being a reference for representing a relative height of a keypoint. The reference point may be preset, or may be able to be selected by a user. The reference point is preferably at the center of the skeleton structure or higher than the center (in an upper half of an image in the up-down direction), and, for example, coordinates of a keypoint of a neck are set as the reference point. Note that coordinates of a keypoint of a head or another portion instead of a neck may be set as the reference point. Instead of a keypoint, any coordinates (for example, center coordinates in the skeleton structure, and the like) may be set as the reference point.

Subsequently, the feature value computation unit 103 normalizes the keypoint height (yi) by the height pixel count

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(S243). The feature value computation unit 103 normalizes each keypoint by using the keypoint height of each keypoint, the reference point, and the height pixel count. Specifically, the feature value computation unit 103 normalizes, by the height pixel count, a relative height of a keypoint with respect to the reference point. Herein, as an example focusing only on the height direction, only a Y coordinate is extracted, and normalization is performed with the reference point as the keypoint of the neck. Specifically, with a Y coordinate of the reference point (keypoint of the neck) as (yc), a feature value (normalization value) is acquired by using the following equation (1). Note that, when a vertical projection axis based on a camera parameter is used, (yi) and (yc) are converted to values in a direction along the vertical projection axis.

[Mathematical 1]

$$f_i = (y_i - y_c) / h \quad (1)$$

For example, when 18 keypoints are present, 18 coordinates (x0, y0), (x1, y1), . . . and (x17, y17) of the keypoints are converted to 18-dimensional feature values as follows by using the equation (1) described above.

[Mathematical 2]

$$\begin{aligned} f_0 &= (y_0 - y_c) / h \\ f_1 &= (y_1 - y_c) / h \\ &\dots \\ f_{17} &= (y_{17} - y_c) / h \end{aligned} \quad (2)$$

FIG. 36 illustrates an example of a feature value of each keypoint acquired by the feature value computation unit 103. In this example, since the keypoint A2 of the neck is the reference point, a feature value of the keypoint A2 is 0.0 and a feature value of a keypoint A31 of a right shoulder and a keypoint A32 of a left shoulder at the same height as the neck is also 0.0. A feature value of a keypoint A1 of a head higher than the neck is -0.2. Feature values of a keypoint A51 of a right hand and a keypoint A52 of a left hand lower than the neck are 0.4, and feature values of the keypoint A81 of the right foot and the keypoint A82 of the left foot are 0.9. When the person raises the left hand from this state, the left hand is higher than the reference point as in FIG. 37, and thus a feature value of the keypoint A52 of the left hand is -0.4. Meanwhile, since normalization is performed by using only a coordinate of the Y axis, as in FIG. 38, a feature value does not change as compared to FIG. 36 even when a width of the skeleton structure changes. In other words, a feature value (normalization value) according to the present example embodiment indicates a feature of a skeleton structure (keypoint) in the height direction (Y direction), and is not affected by a change of the skeleton structure in the horizontal direction (X direction).

As described above, in the present example embodiment, a skeleton structure of a person is detected from a two-dimensional image, and each keypoint of the skeleton structure is normalized by using a height pixel count (upright height on a two-dimensional image space) acquired from the detected skeleton structure. Robustness when classification, a search, and the like are performed can be improved by using the normalized feature value. In other words, since a feature value according to the present example embodiment is not affected by a change of a person in the horizontal

direction as described above, robustness with respect to a change in orientation of the person and a body shape of the person is great.

Furthermore, the present example embodiment can be achieved by detecting a skeleton structure of a person by using a skeleton estimation technique such as Open Pose, and thus learning data that learn a pose and the like of a person do not need to be prepared. Further, classification and a search of a pose and the like of a person can be achieved by normalizing a keypoint of a skeleton structure and storing the keypoint in advance in a database, and thus classification and a search can also be performed on an unknown pose. Further, a clear and simple feature value can be acquired by normalizing a keypoint of a skeleton structure, and thus persuasion of a user for a processing result is high unlike a black box algorithm as in machine learning.

While the example embodiments of the present invention have been described with reference to the drawings, the example embodiments are only exemplification of the present invention, and various configurations other than the above-described example embodiments can also be employed.

Further, the plurality of steps (pieces of processing) are described in order in the plurality of flowcharts used in the above-described description, but an execution order of steps performed in each of the example embodiments is not limited to the described order. In each of the example embodiments, an order of illustrated steps may be changed within an extent that there is no harm in context. Further, each of the example embodiments described above can be combined within an extent that a content is not inconsistent.

A part or the whole of the above-described example embodiment may also be described in supplementary notes below, which is not limited thereto.

1. An image selection apparatus, including:

- a query acquisition unit that acquires a query video including a plurality of first frame images;
- a query frame selection unit that selects a plurality of query frames from among the plurality of first frame images; and
- a video selection unit that selects a video by using the plurality of query frames, in which the video selection unit uses at least (1) and (2) below as a condition for selecting the video.

(1) A condition that a “similar frame image whose degree of similarity to the query frame satisfies a first reference is present” is satisfied for at least two of the query frames.

(2) When the query frames associated with each of the at least two similar frame images are arranged in a same order as that of the at least two similar frame images, the arrangement order coincides with an arrangement order of the at least two query frames in the query video.

2. The image selection apparatus according to supplementary note 1 described above, in which

- the query video includes a person, and
- the video selection unit uses, as the degree of similarity, a degree of similarity of a pose of the person.

3. The image selection apparatus according to supplementary note 2 described above, in which

- one of conditions needed to be satisfied by the query frame is that an information amount related to the person included in the query frame satisfies a second reference.

4. The image selection apparatus according to supplementary note 3 described above, in which

each of the plurality of first frame images includes, as information indicating a pose of the person, joint information indicating a position of a joint of the person, and the second reference is that a number of joints included in the joint information satisfies a reference.

5. The image selection apparatus according to any one of supplementary notes 1 to 4 described above, in which the query frame selection unit selects at least one of the query frames according to an input from a user.

6. The image selection apparatus according to any one of supplementary notes 1 to 4 described above, in which the query frame selection unit repeats processing of selecting, as the query frame following a first of the query frames, the first frame image having a change amount from the first query frame equal to or more than a third reference.

7. The image selection apparatus according to supplementary note 6 described above, in which

- the query frame selection unit sets the third reference by using an input from a user.

8. The image selection apparatus according to supplementary note 6 described above, in which

- the query frame selection unit sets the third reference used for the query frame by using the plurality of first frame images.

9. The image selection apparatus according to any one of supplementary notes 1 to 4 described above, in which

- the query frame selection unit selects a plurality of query frames from among the plurality of first frame images at predetermined intervals.

10. The image selection apparatus according to any one of supplementary notes 6 to 9 described above, in which

- the query frame selection unit selects an n-th of the first frame images as a first of the query frames, in which n is an integer set in advance or by a user input.

11. An image selection method, including,

- by a computer:
- executing acquisition processing of acquiring a query video including a plurality of first frame images;
- executing query frame selection processing of selecting a plurality of query frames from among the plurality of first frame images;
- executing video selection processing of selecting a video by using the plurality of query frames; and,
- in the video selection processing, using at least (1) and (2) below as a condition for selecting the video.

(1) A condition that a “similar frame image whose degree of similarity to the query frame satisfies a first reference is present” is satisfied for at least two of the query frames.

(2) When the query frames associated with each of the at least two similar frame images are arranged in a same order as that of the at least two similar frame images, the arrangement order coincides with an arrangement order of the at least two query frames in the query video.

12. The image selection method according to supplementary note 11 described above, in which

- the query video includes a person,
- the image selection method further including,
- by the computer,
- in the video selection processing, using a degree of similarity of a pose of the person as the degree of similarity.

13. The image selection method according to supplementary note 12 described above, in which

- one of conditions needed to be satisfied by the query frame is that an information amount related to the person included in the query frame satisfies a second reference.
14. The image selection method according to supplementary note 13 described above, in which
- each of the plurality of first frame images includes, as information indicating a pose of the person, joint information indicating a position of a joint of the person, and the second reference is that a number of joints included in the joint information satisfies a reference.
15. The image selection method according to any one of supplementary notes 11 to 14 described above, further including,
- by the computer,
 - in the query frame selection processing, selecting at least one of the query frames according to an input from a user.
16. The image selection method according to any one of supplementary notes 11 to 14 described above, further including,
- by the computer,
 - in the query frame selection processing, repeating processing of selecting, as the query frame following a first of the query frames, the first frame image having a change amount from the first query frame equal to or more than a third reference.
17. The image selection method according to supplementary note 16 described above, further including,
- by the computer,
 - in the query frame selection processing, setting the third reference by using an input from a user.
18. The image selection method according to supplementary note 16 described above, further including,
- by the computer,
 - in the query frame selection processing, setting the third reference used for the query frame by using the plurality of first frame images.
19. The image selection method according to any one of supplementary notes 11 to 14 described above, further including,
- by the computer,
 - in the query frame selection processing, selecting a plurality of query frames from among the plurality of first frame images at predetermined intervals.
20. The image selection method according to any one of supplementary notes 16 to 19 described above, further including,
- by the computer,
 - in the query frame selection processing, selecting an n-th of the first frame images as a first of the query frames, in which
 - n is an integer set in advance or by a user input.
21. A program causing a computer to include:
- an acquisition function of acquiring a query video including a plurality of first frame images;
 - a query frame selection function of selecting a plurality of query frames from among the plurality of first frame images; and
 - a video selection function of selecting a video by using the plurality of query frames, in which
 - the video selection function uses at least (1) and (2) below as a condition for selecting the video.
- (1) A condition that a “similar frame image whose degree of similarity to the query frame satisfies a first reference is present” is satisfied for at least two of the query frames.

- (2) When the query frames associated with each of the at least two similar frame images are arranged in a same order as that of the at least two similar frame images, the arrangement order coincides with an arrangement order of the at least two query frames in the query video.
22. The program according to supplementary note 21 described above, in which
- the query video includes a person, and
 - the video selection function uses a degree of similarity of a pose of the person as the degree of similarity.
23. The program according to supplementary note 22 described above, in which
- one of conditions needed to be satisfied by the query frame is that an information amount related to the person included in the query frame satisfies a second reference.
24. The program according to supplementary note 23 described above, in which
- each of the plurality of first frame images includes, as information indicating a pose of the person, joint information indicating a position of a joint of the person, and the second reference is that a number of joints included in the joint information satisfies a reference.
25. The program according to any one of supplementary notes 21 to 24 described above, in which
- the query frame selection function selects at least one of the query frames according to an input from a user.
26. The program according to any one of supplementary notes 21 to 24 described above, in which
- the query frame selection function repeats processing of selecting, as the query frame following a first of the query frames, the first frame image having a change amount from the first query frame equal to or more than a third reference.
27. The program according to supplementary note 26 described above, in which
- the query frame selection function sets the third reference by using an input from a user.
28. The program according to supplementary note 26 described above, in which
- the query frame selection function sets the third reference used for the query frame by using the plurality of first frame images.
29. The program according to any one of supplementary notes 21 to 24 described above, in which
- the query frame selection function selects a plurality of query frames from among the plurality of first frame images at predetermined intervals.
30. The program according to any one of supplementary notes 26 to 29 described above, in which
- the query frame selection function selects an n-th of the first frame images as a first of the query frames, in which
 - n is an integer set in advance or by a user input.

REFERENCE SIGNS LIST

- 1 Image processing system
- 10 Image processing apparatus (image selection apparatus)
- 11 Skeleton detection unit
- 12 Feature value computation unit
- 13 Recognition unit
- 100 Image processing apparatus (image selection apparatus)
- 101 Image acquisition unit
- 102 Skeleton structure detection unit

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103 Feature value computation unit

104 Classification unit

105 Search unit

106 Input unit

107 Display unit

108 Height computation unit

110 Database

200 Camera

300, 301 Human model

401 Two-dimensional skeleton structure

610 Query acquisition unit

620 Query frame selection unit

630 Video selection unit

What is claimed is:

1. An image selection apparatus comprising:

at least one memory storing instructions; and

at least one processor configured to execute the instructions to perform operations comprising:

acquiring a query video including a plurality of first frame images;

selecting a plurality of query frames from among the plurality of first frame images, wherein the plurality of query frames are selected by repeatedly performing a process of selecting, as a query frame following a first of the query frames, the first frame image having a change amount from the first query frame equal to or greater than a predetermined reference; and

selecting a video by using the plurality of query frames, based on at least first and second conditions for selecting the video, wherein

the first condition is that a similar frame image having a degree of similarity to the query frame satisfies a first reference is present for at least two of the query frames,

the second condition is that, in a case in which the query frames associated with each of at least two similar frame images are arranged in a same order as that of the at least two similar frame images, an arrangement order of the at least two similar frame images coincides with an arrangement order of the at least two query frames in the query video.

2. The image selection apparatus according to claim 1, wherein

the query video includes a person, and

the operations comprise using, as the degree of similarity, a degree of similarity of a pose of the person.

3. The image selection apparatus according to claim 2, wherein

a third condition that is to be satisfied by the query frame is that an information amount related to the person included in the query frame satisfies a second reference.

4. The image selection apparatus according to claim 3, wherein

each of the plurality of first frame images includes, as information indicating a pose of the person, joint information indicating a position of a joint of the person, and the second reference is that a number of joints included in the joint information satisfies a reference.

5. The image selection apparatus according to claim 1, wherein

the operations comprise selecting at least one of the query frames according to an input from a user.

6. The image selection apparatus according to claim 1, wherein

the operations comprise setting the predetermined reference by using an input from a user.

7. The image selection apparatus according to claim 1, wherein

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the operations comprise setting the third predetermined reference by using the plurality of first frame images.

8. The image selection apparatus according to claim 1, wherein

the operations comprise selecting the plurality of query frames from among the plurality of first frame images at predetermined intervals.

9. The image selection apparatus according to claim 1, wherein

the operations comprise selecting an n-th of the first frame images as a first of the query frames, and n is an integer set in advance or by a user input.

10. An image selection method performed by a computer and comprising:

by a computer:

acquiring a query video including a plurality of first frame images;

selecting a plurality of query frames from among the plurality of first frame images, wherein the plurality of query frames are selected by repeating processing of selecting, as a query frame following a first of the query frames, the first frame image having a change amount from the first query frame equal to or more than a predetermined reference; and

selecting a video by using the plurality of query frames, based on at least first and second conditions selecting the video, wherein

the first condition is that a similar frame image having a degree of similarity to the query frame satisfies a first reference is present is satisfied for at least two of the query frames,

the second condition is that, in a case in which the query frames associated with each of at least two similar frame images are arranged in a same order as that of the at least two similar frame images, an arrangement order of the at least two similar frame images coincides with an arrangement order of the at least two query frames in the query video.

11. A non-transitory computer-readable medium storing a program executable by a computer to perform operations comprising:

acquiring a query video including a plurality of first frame images;

selecting a plurality of query frames from among the plurality of first frame images, wherein the plurality of query frames are selected by repeating processing of selecting, as a query frame following a first of the query frames, the first frame image having a change amount from the first query frame equal to or more than a predetermined reference; and

selecting a video by using the plurality of query frames, based on at least first and second conditions selecting the video, wherein

the first condition is that a similar frame image having a degree of similarity to the query frame satisfies a first reference is present is satisfied for at least two of the query frames,

the second condition is that, in a case in which the query frames associated with each of at least two similar frame images are arranged in a same order as that of the at least two similar frame images, an arrangement order of the at least two similar frame images coincides with an arrangement order of the at least two query frames in the query video.